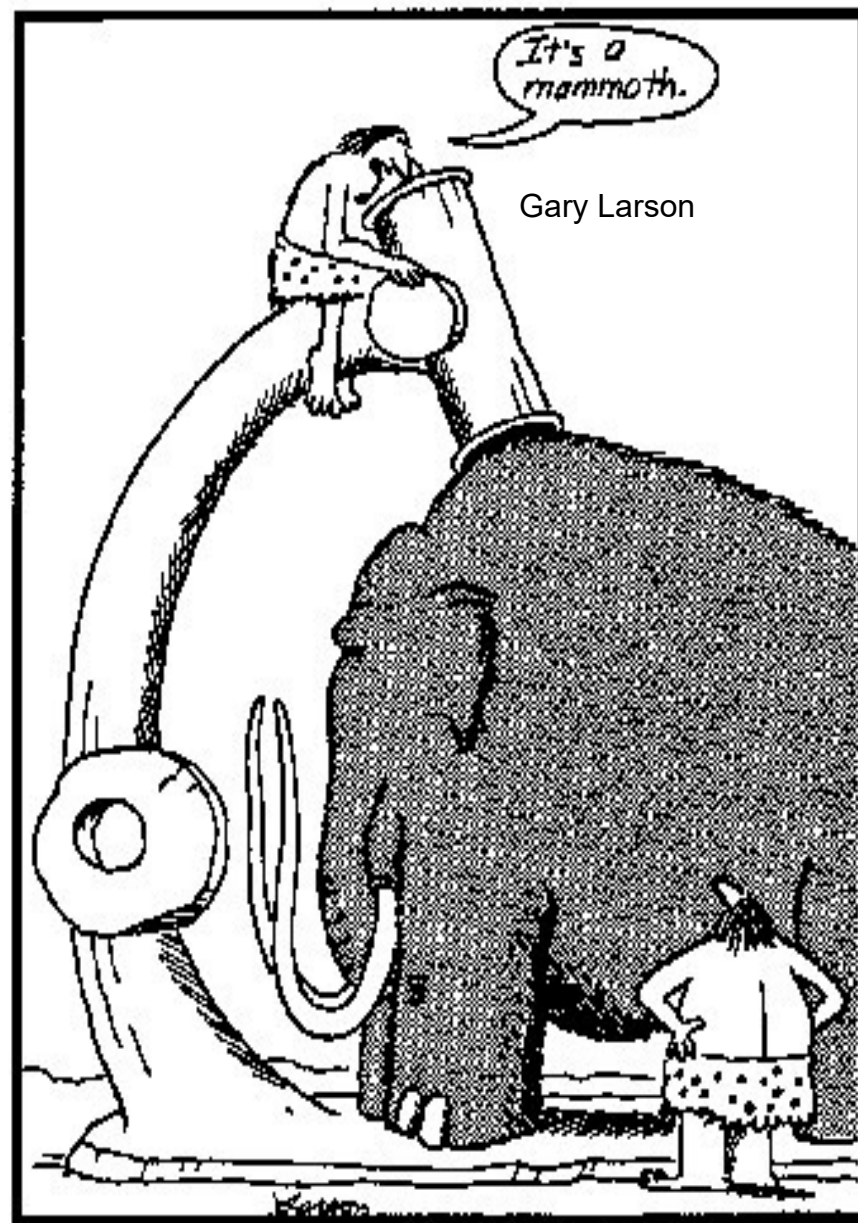


Geometrical optics

- refraction, Snell's law
- total reflection
- lenses, thin lens equation
- imaging, microscope, telescope
- eyesight.





Gary Larson

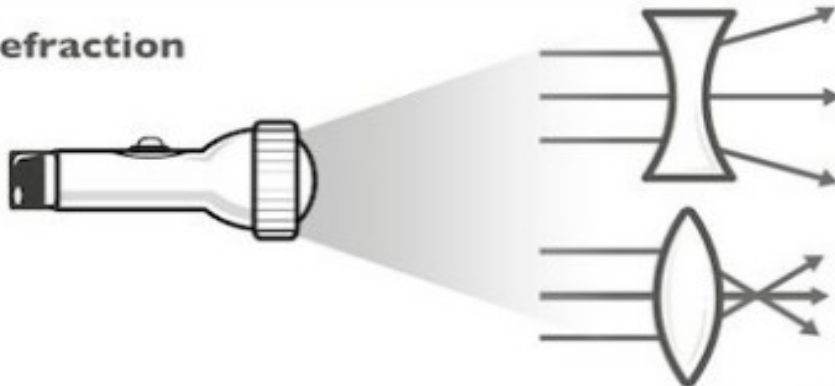
Early microscope

Reflection & Refraction

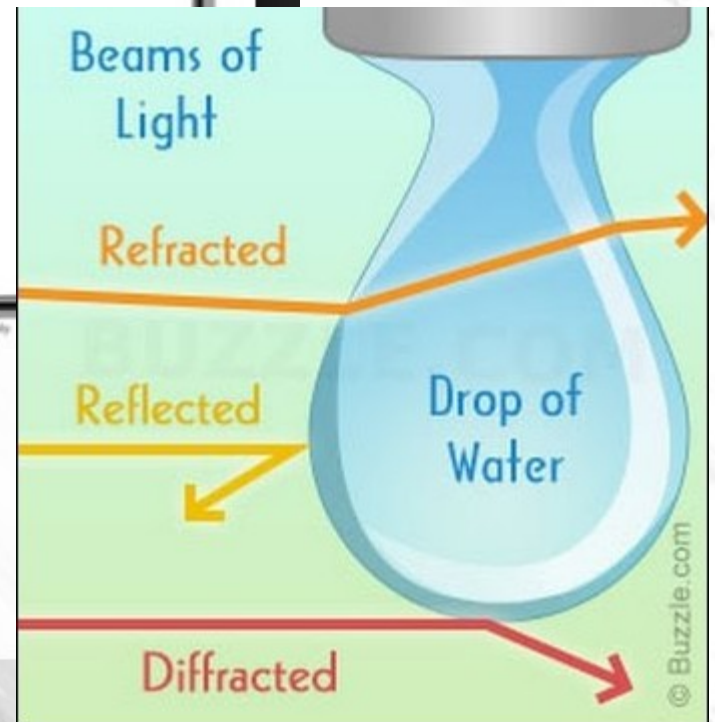
Reflection



Refraction



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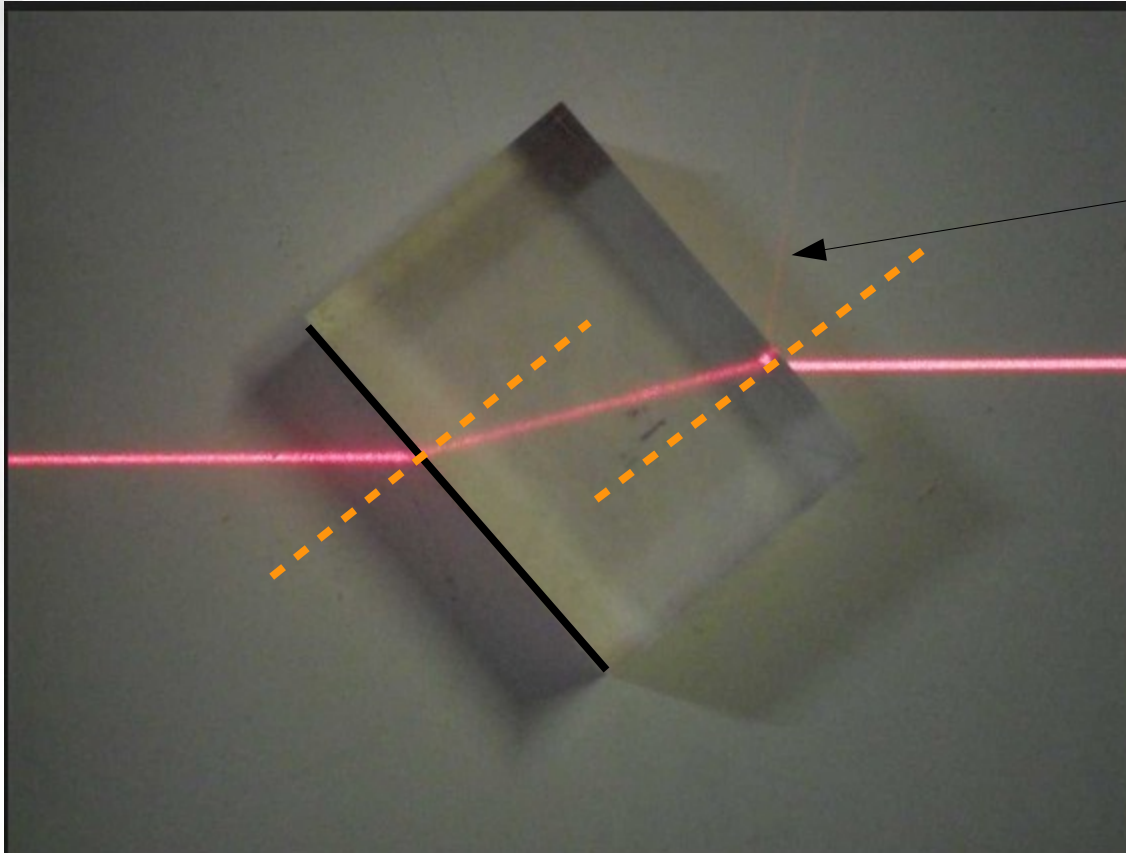


REFRACTION of waves. Bending of waves

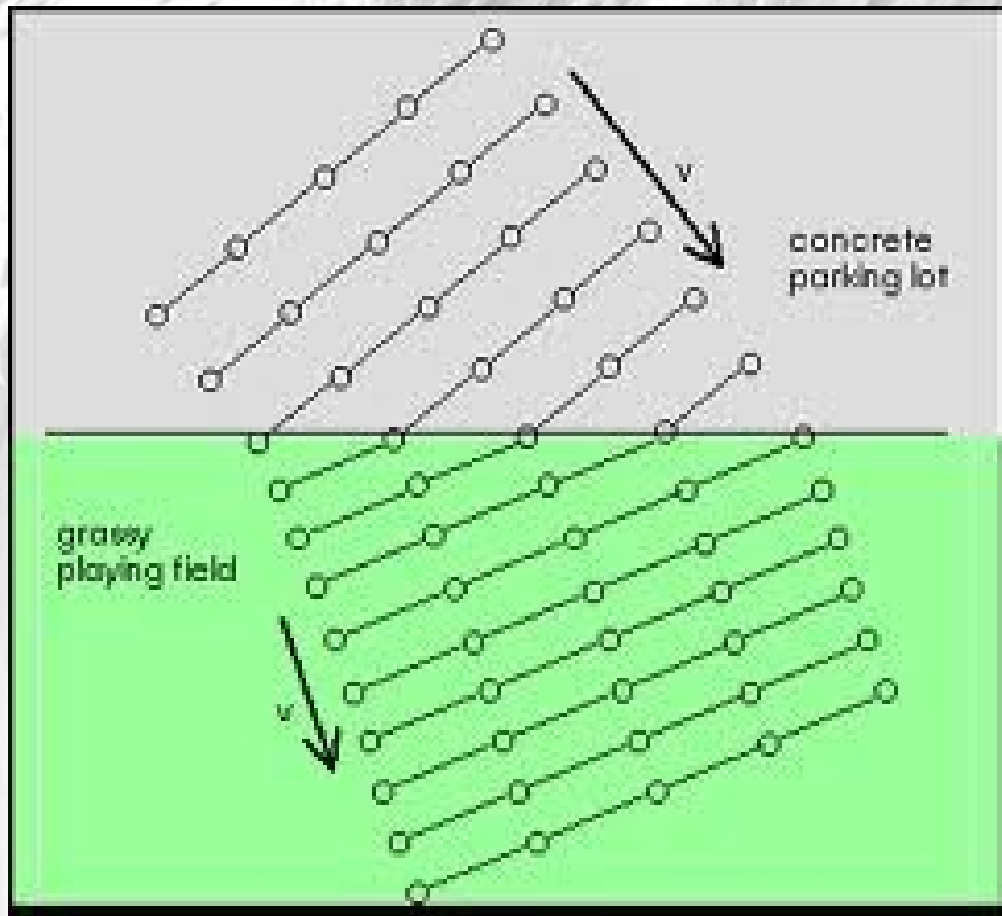
Refraction: the change in direction of a wave passing from one medium to another caused by its change in speed.

Waves bend toward the region of slow speed. Like a marching band
Going from grass to Mud bend. Sound bends, light bends.

https://phet.colorado.edu/sims/html/bending-light/latest/bending-light_en.html



**The orange line
Is called the normal
To the surface
That separates the
2 media**



Rays are perpendicular to Wave fronts.

See Movie

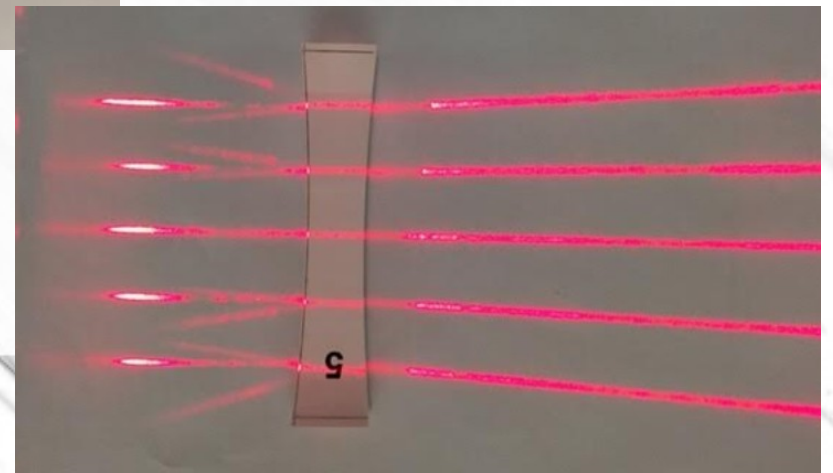
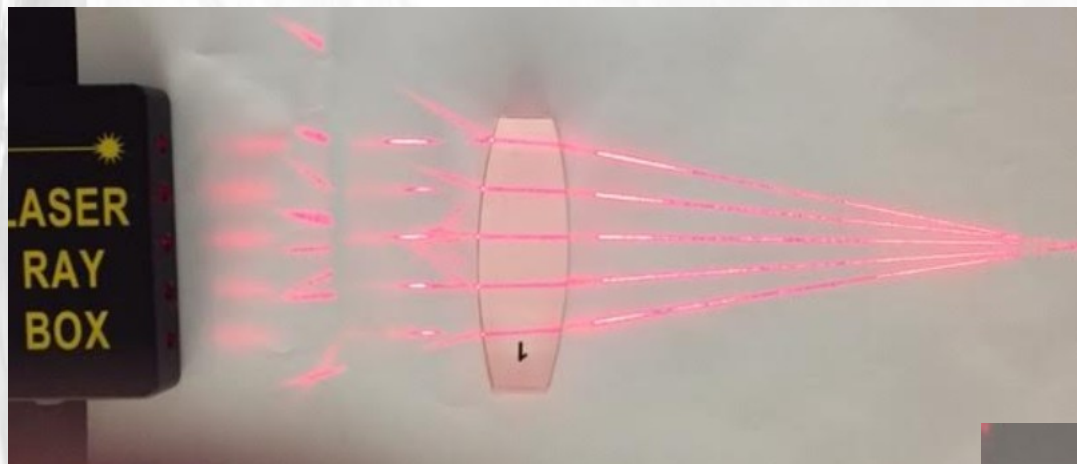
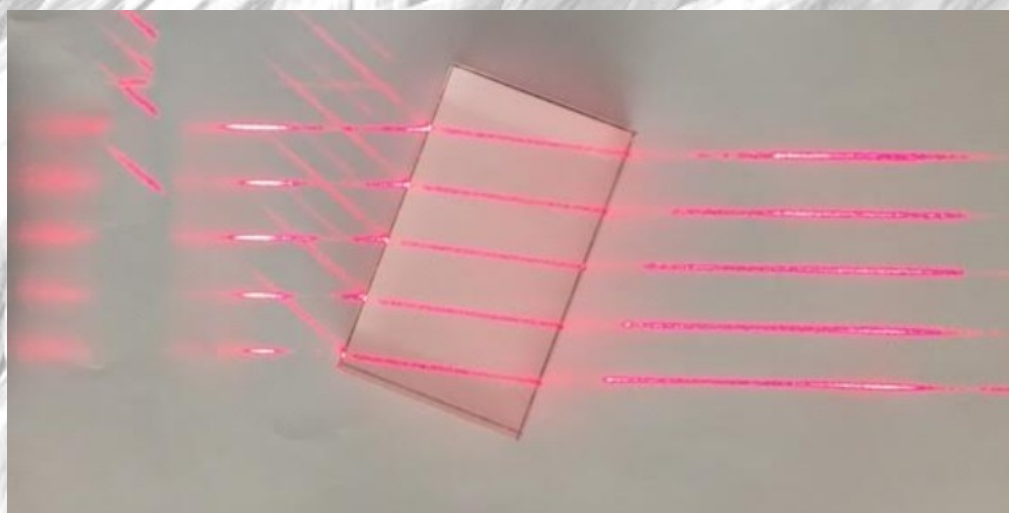
$\frac{c}{v}$ (see link) is called the index of refraction of the piece of material.

<https://phet.colorado.edu/en/simulations/bending-light>

5

The speed of light is smaller so the wavelengths get smaller

- When light travels from air into water
 - A. its velocity, wavelength, and frequency all change.
 - B. its velocity changes, but its frequency and wavelength do not change.
 - C. its frequency changes, but its velocity and wavelength do not change.
 - D. its velocity and wavelength change, but its frequency does not change.
 - E. its wavelength changes, but its velocity and frequency do not change.



SOUND WAVES bend

Sound travels faster in warm air (because the air molecules jiggle faster)
Than in cold air. (less kinetic energy = less speed)

This phenomenon can explain why in the morning you
Can't hear far away sounds. The air above the ground is warm and
The upper air is cold.

Sound missed me



Draw the path of sound waves emitted by the
Dog. Do they travel in a straight line ?

Cooler and cooler (layers)
(SLOW)

Warmer and warmer
(FAST)



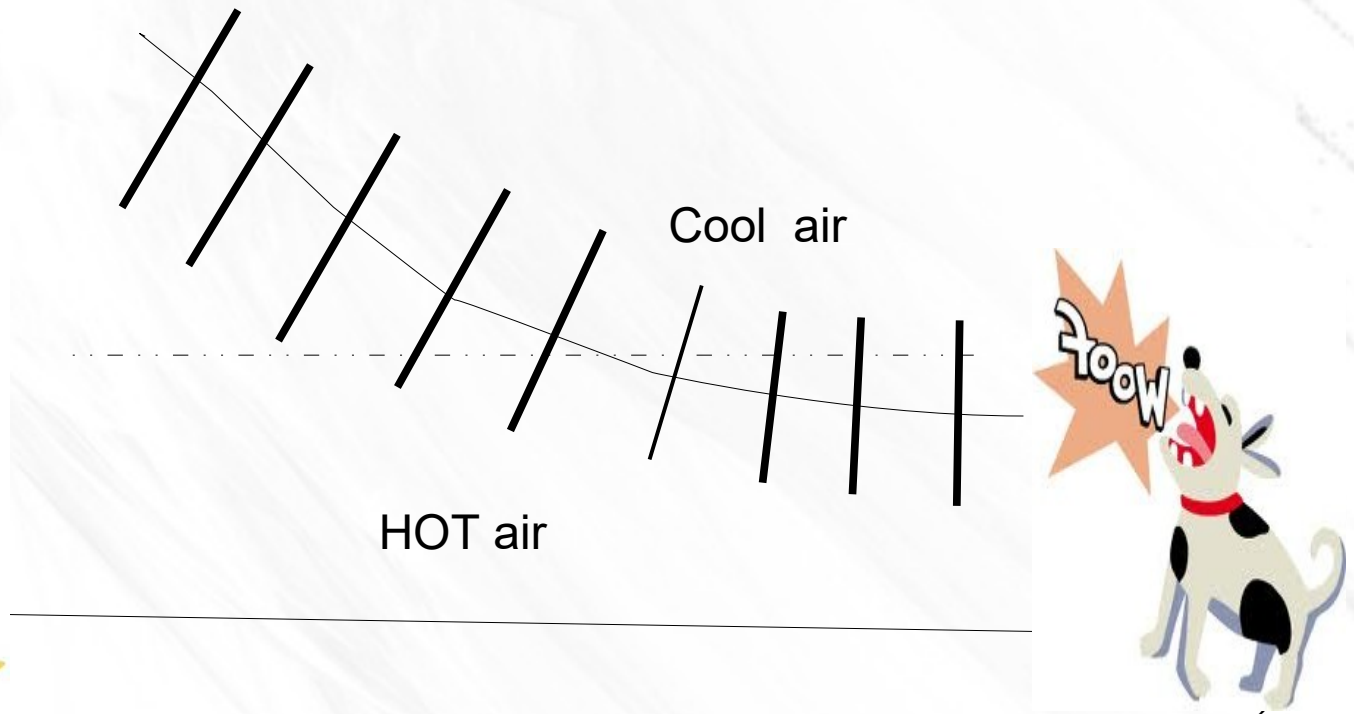
Note: the speed of sound = speed of molecules.
Temperature increases = faster motion

SOUND WAVES bend

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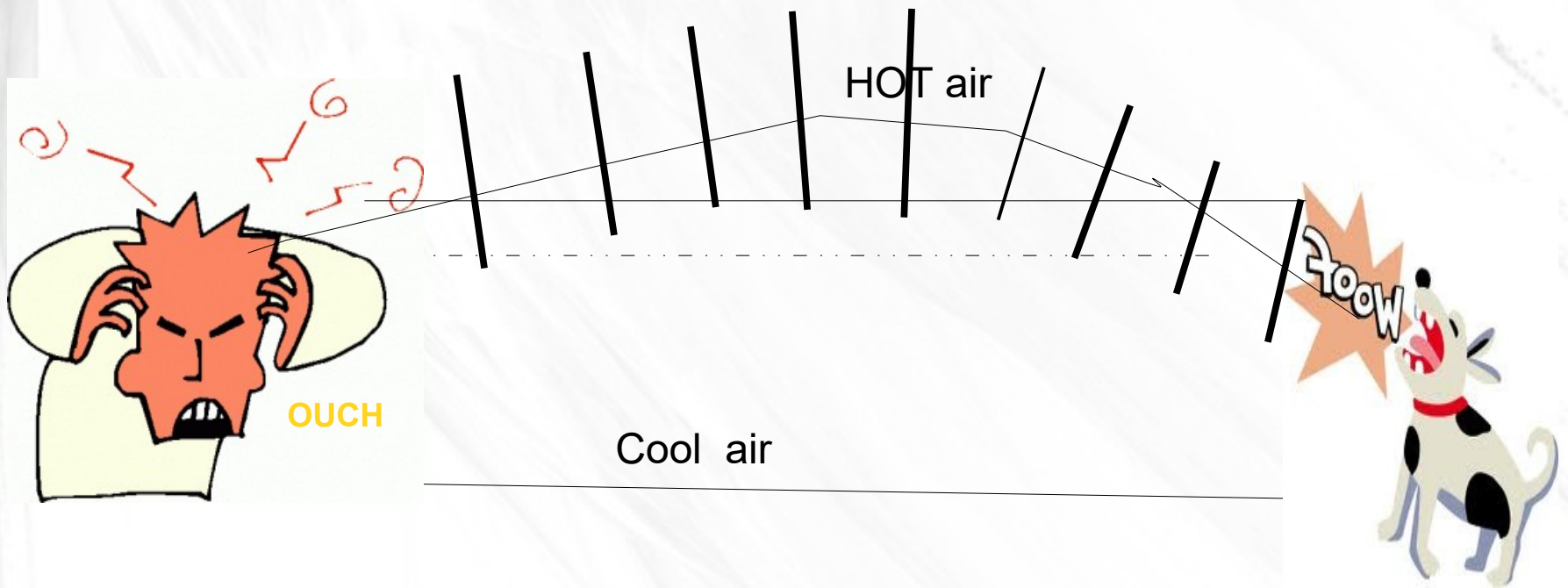
Sound missed me



Note: the speed of sound = speed of molecules.
Temperature increases = faster motion

SOUND WAVES bend

In the evening it is the opposite. The air is cooler above the ground
So you can hear far away sounds



SOUND WAVES bend

Sometimes during the day an inversion can happen

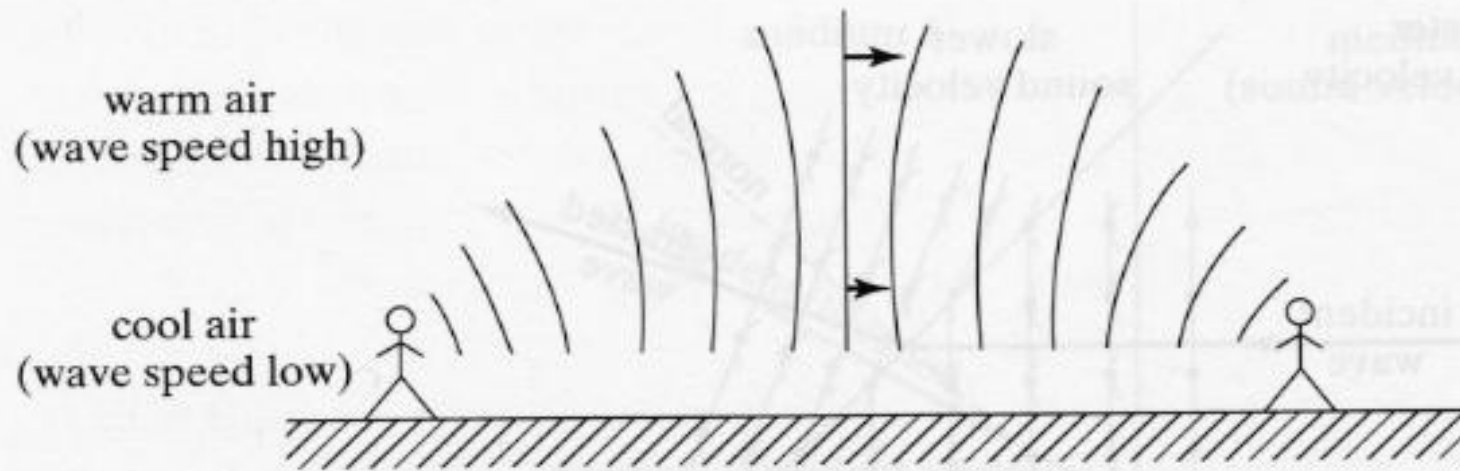
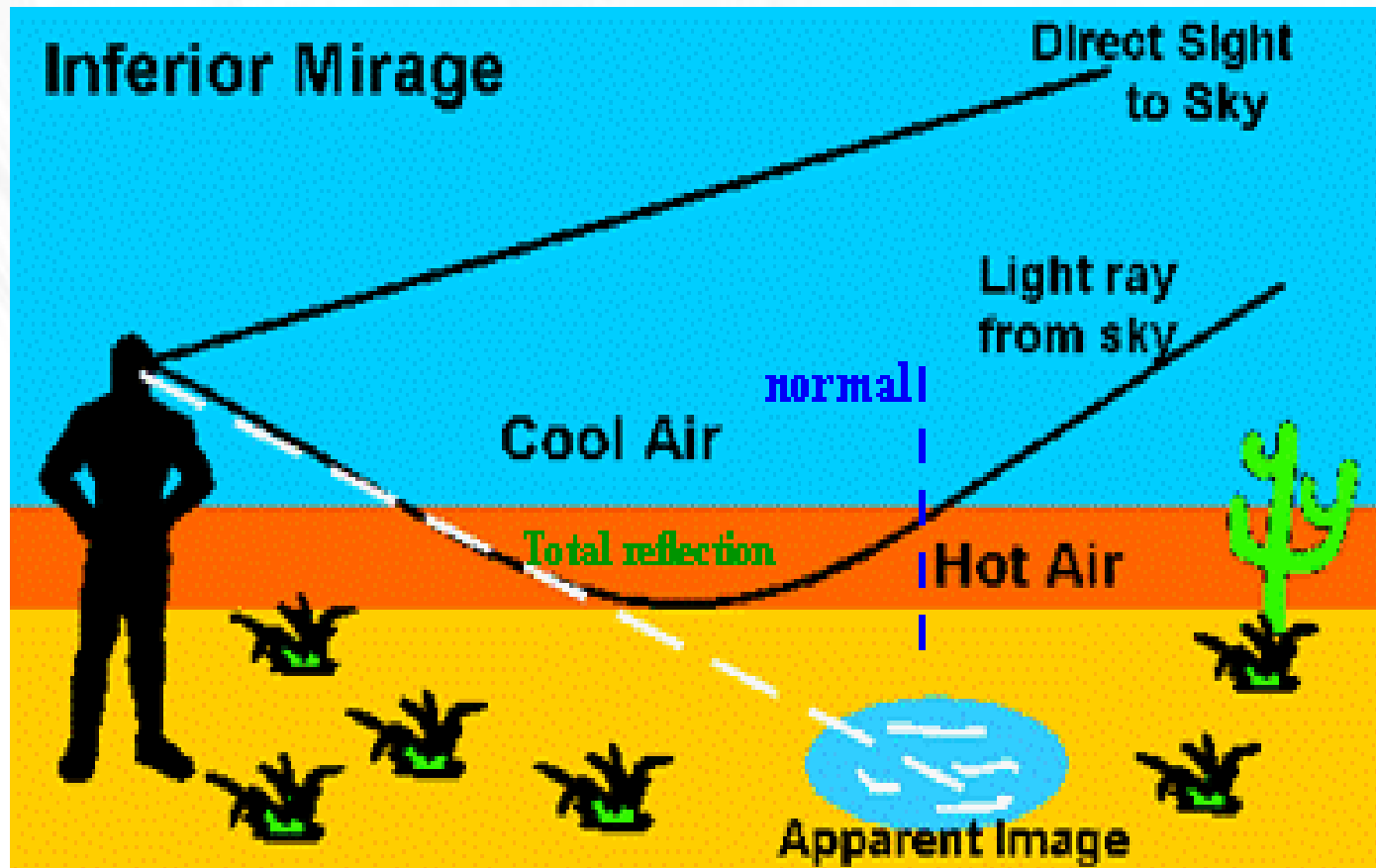


Figure 2-22 During a temperature inversion, sound waves emitted upward are refracted back toward the earth, causing the sound to travel much further along the ground.

Light waves can bend

This bending of light based on layer of different temperature
Also explain the mirages. In that case light is bent.

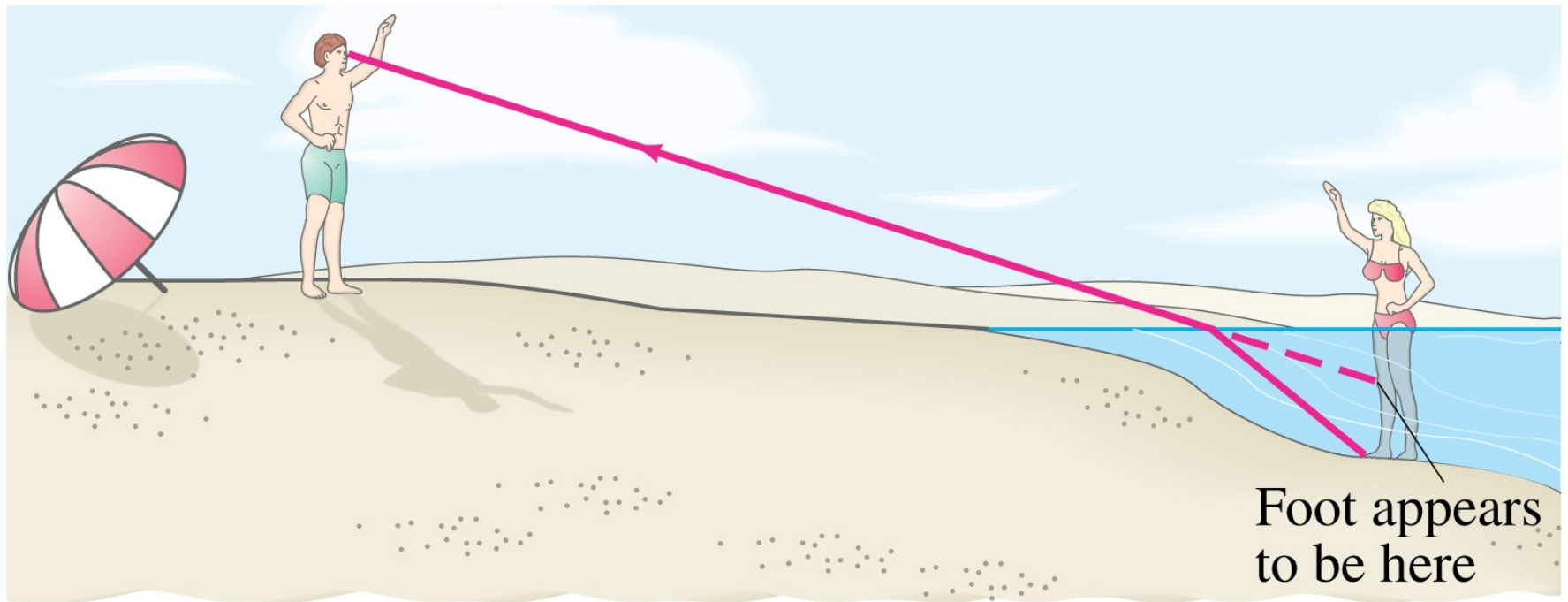
FROM SLOW TO FAST = AWAY . In that case, so much that there is Total reflection



23.5 Refraction: Snell's Law

Refraction is what makes objects half-submerged in water look odd.

<https://phet.colorado.edu/en/simulation/bending-light>



Light waves can bend

Light travels slower in water than in air.

When the hunter looks at the fish, the

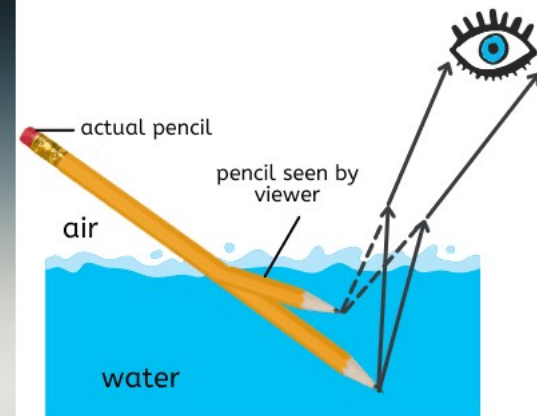
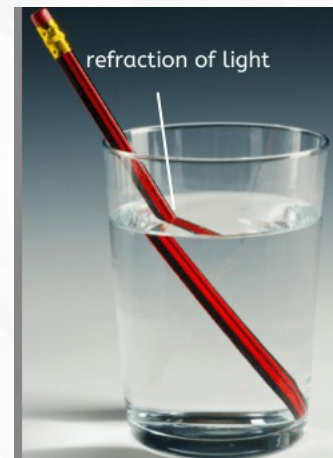
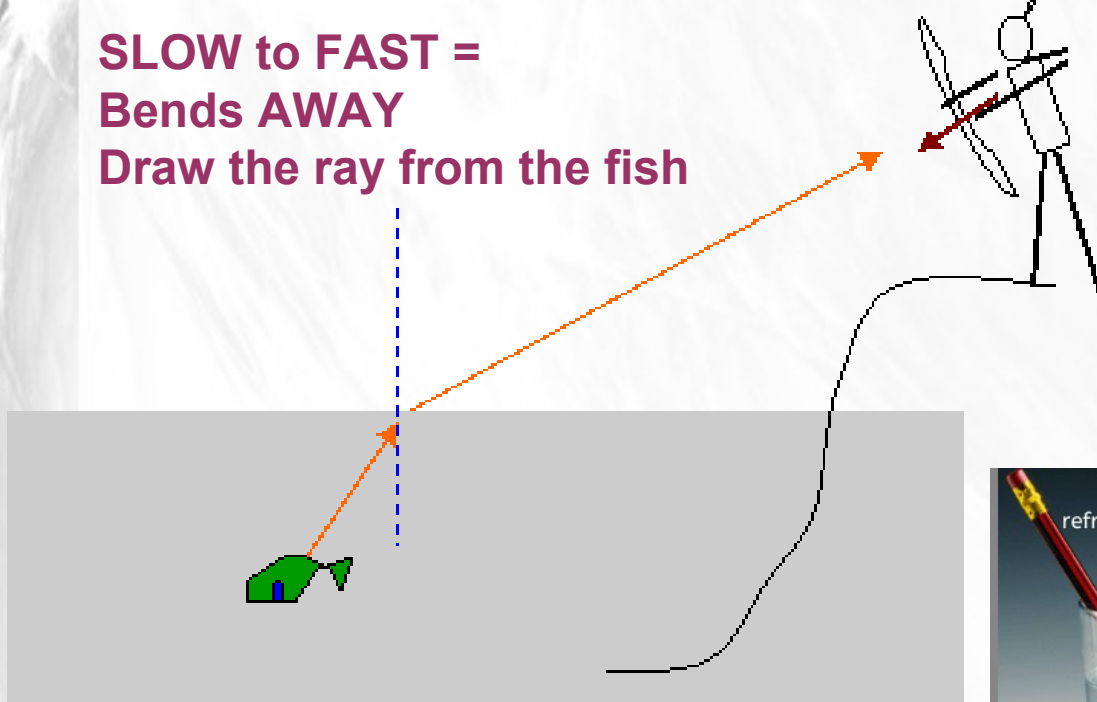
Light coming from the fish bends when crossing the boundary water-air.

Where is the image of the fish? Will he/she kill the fish?

SLOW to FAST =

Bends AWAY

Draw the ray from the fish



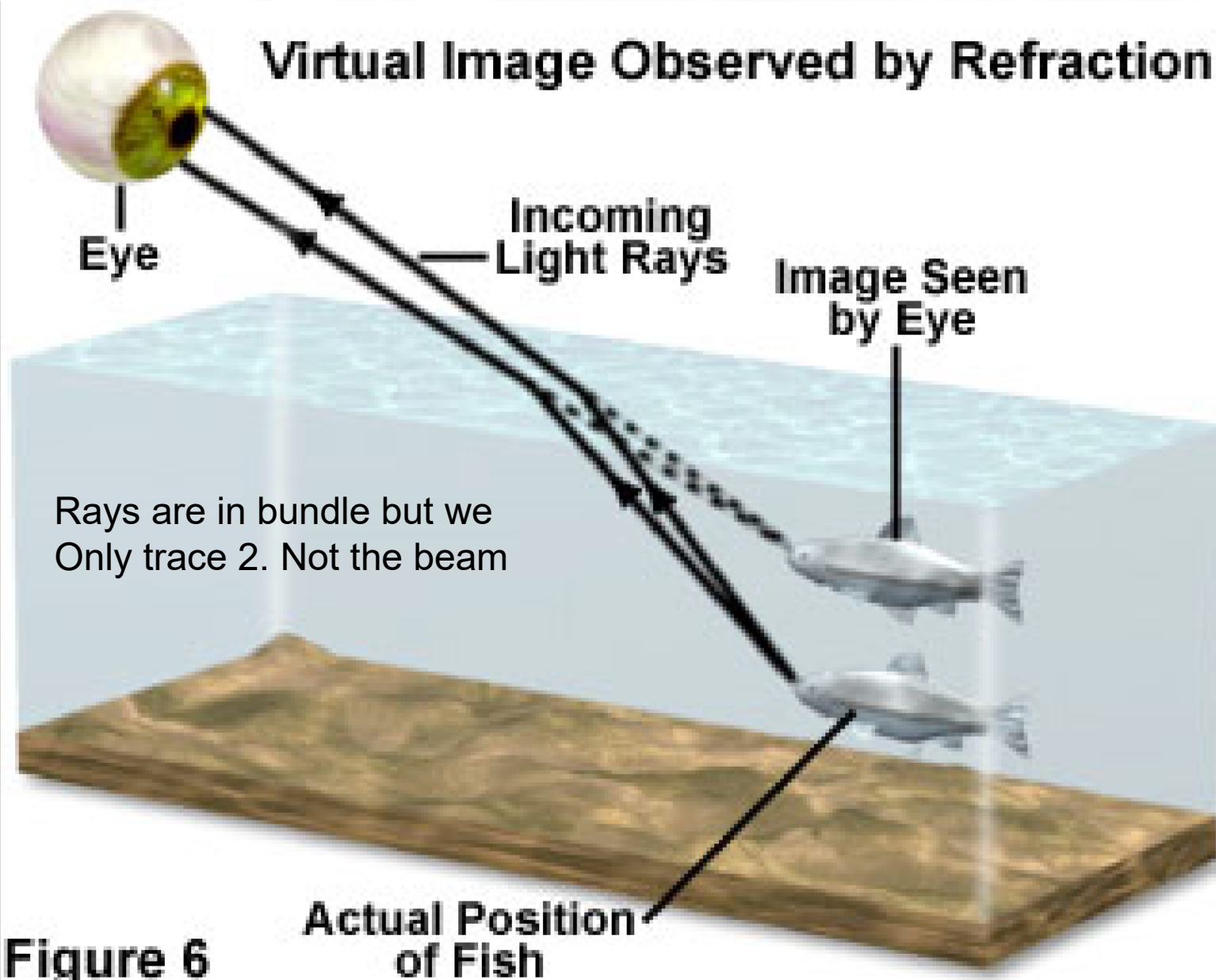


Figure 6

26.1 The Index of Refraction

<http://phet.colorado.edu/en/simulation/bending-light>
Discuss the app. What happens when you increase the density of medium 2 ? Discuss the % reflected Vs refracted. Discuss the speed. Change to waves. See if beam bends toward or away.

DEFINITION OF THE INDEX OF REFRACTION

The index of refraction of a material is the ratio of the speed of light in a vacuum to the speed of light in the material:

$$n = \frac{\text{Speed of light in vacuum}}{\text{Speed of light in the material}} = \frac{c}{v}$$

Which substance has the highest Index of refraction?

Table 26.1 Index of Refraction^a for Various Substances

Substance	Index of Refraction, n
<i>Solids at 20 °C</i>	
Diamond	2.419
Glass, crown	1.523
Ice (0 °C)	1.309
Sodium chloride	1.544
Quartz	
Crystalline	1.544
Fused	1.458
<i>Liquids at 20 °C</i>	
Benzene	1.501
Carbon disulfide	1.632
Carbon tetrachloride	1.461
Ethyl alcohol	1.362
Water	1.333
<i>Gases at 0 °C, 1 atm</i>	
Air	1.000 293
Carbon dioxide	1.000 45
Oxygen, O ₂	1.000 271
Hydrogen, H ₂	1.000 139

^a Measured with light whose wavelength in a vacuum is 589 nm.

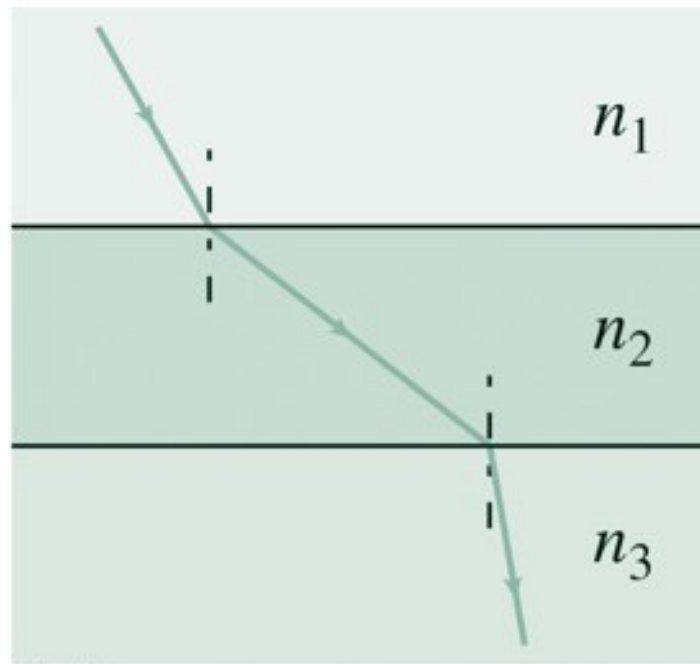
From fast to slow $\rightarrow$ ray moves toward the normal

From slow to fast $\rightarrow$ ray moves away from the normal

$\rightarrow$ larger the index, slower the medium

- The figure shows the path of a light ray through three different media. Rank the media in order of their refractive indices.

- A. $n_1 > n_2 > n_3$
- B. $n_3 > n_1 > n_2$
- C. $n_3 > n_2 > n_1$
- D. $n_2 > n_1 > n_3$



Now some math: refraction of light Or SNELL's LAW

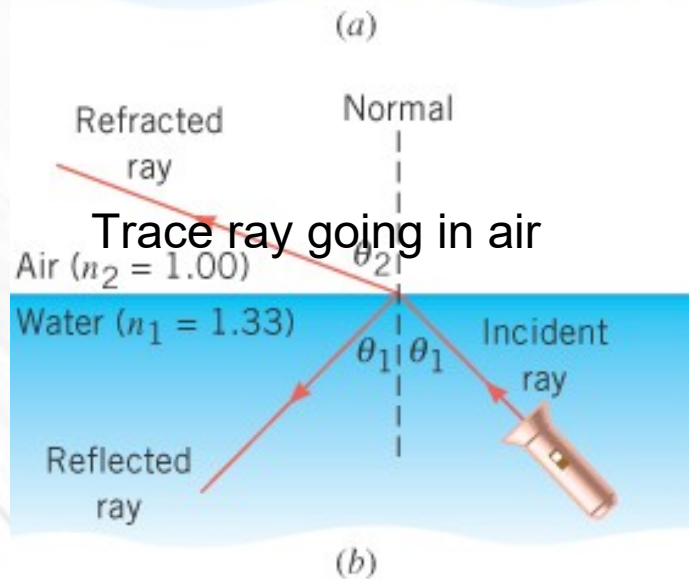
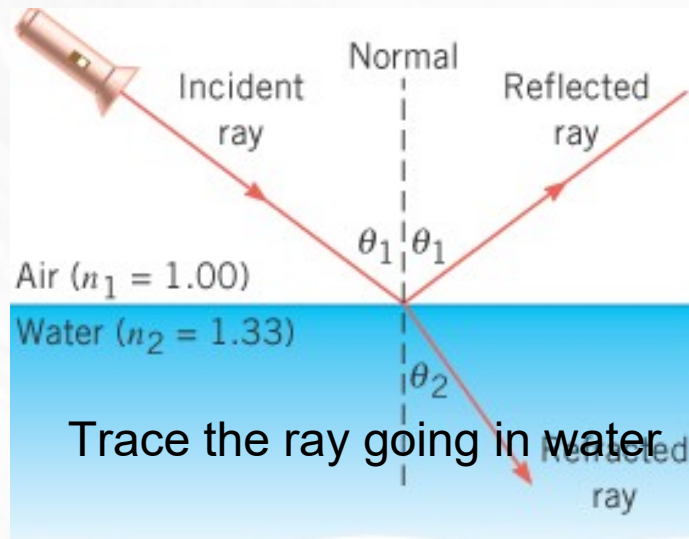
A simple law, called the Snell's law gives amount of Bending of light that cross a boundary between 2 medium With difference speed.

See run exploration
Of physical sciences

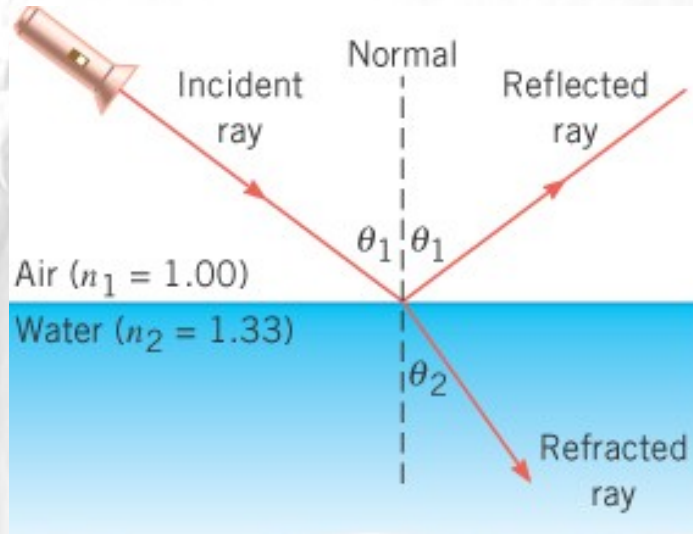
The index n
Depends on
The speed
Of light in
The medium.

$$v = c/n$$

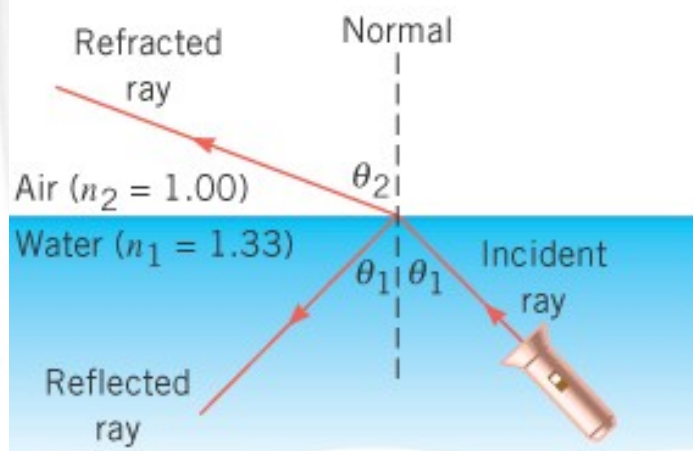
C is the speed
Of light.
 v is the speed
In the medium
 n is the index of
Refraction or
Refractive index



SNELL'S LAW



(a)



(b)

SNELL'S LAW OF REFRACTION

When light travels from a material with one index of refraction to a material with a different index of refraction, the angle of incidence is related to the angle of refraction by

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

GOING from medium 1 (n_1) to medium 2 (n_2)

Refraction: Snell's Law

Wave speed: $v = f\lambda \rightarrow \lambda = v/f$

$$\frac{\lambda_1}{\sin\theta_1} = \frac{\lambda_2}{\sin\theta_2} \rightarrow \frac{\sin\theta_1}{\lambda_1} = \frac{\sin\theta_2}{\lambda_2}$$

Wave crests

Refractive

index: $v = c/n$

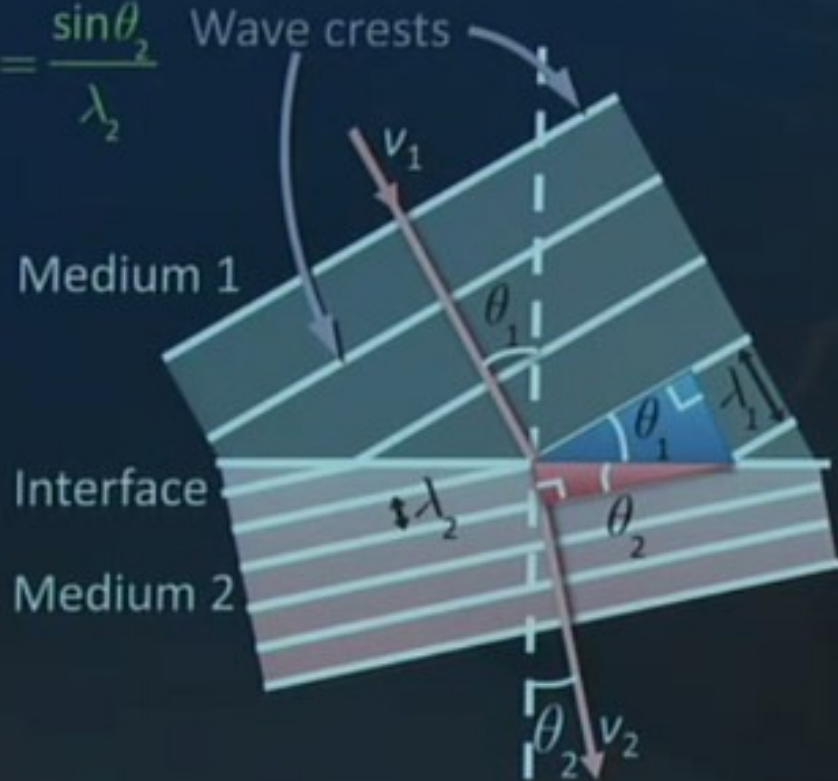
$$\rightarrow \lambda = c/nf$$

So:

$$\frac{n_1}{f} \sin\theta_1 = \frac{n_2}{f} \sin\theta_2$$

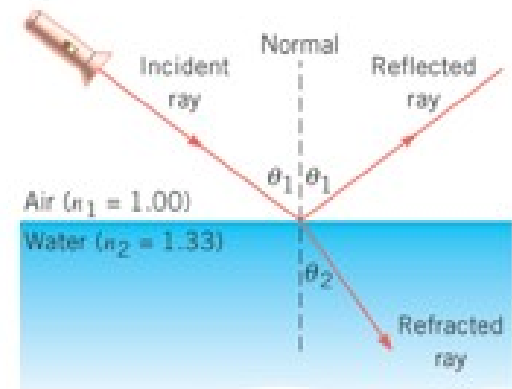
Snell's

law: $n_1 \sin\theta_1 = n_2 \sin\theta_2$

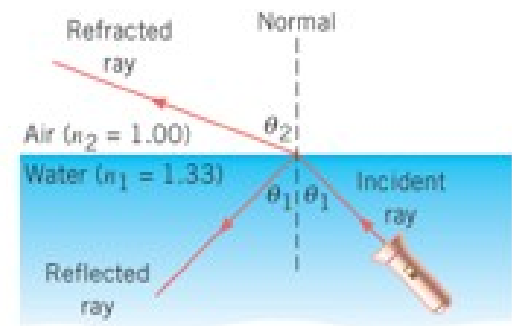


Example 1 Determining the Angle of Refraction

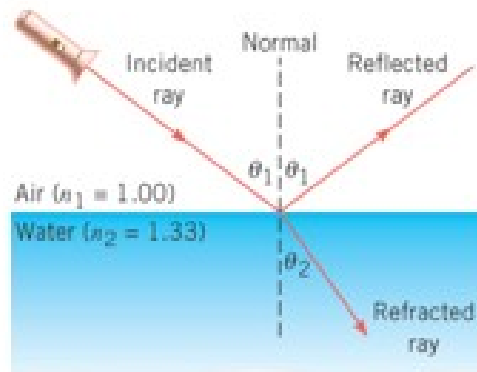
A light ray strikes an air/water surface at an angle of 46 degrees with respect to the normal. Find the angle of refraction when the direction of the ray is (a) from air to water and (b) from water to air.



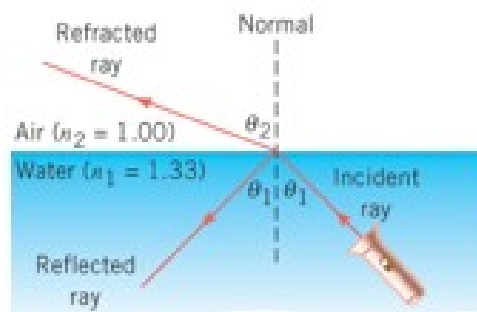
(a)



(b)



(a)



(b)

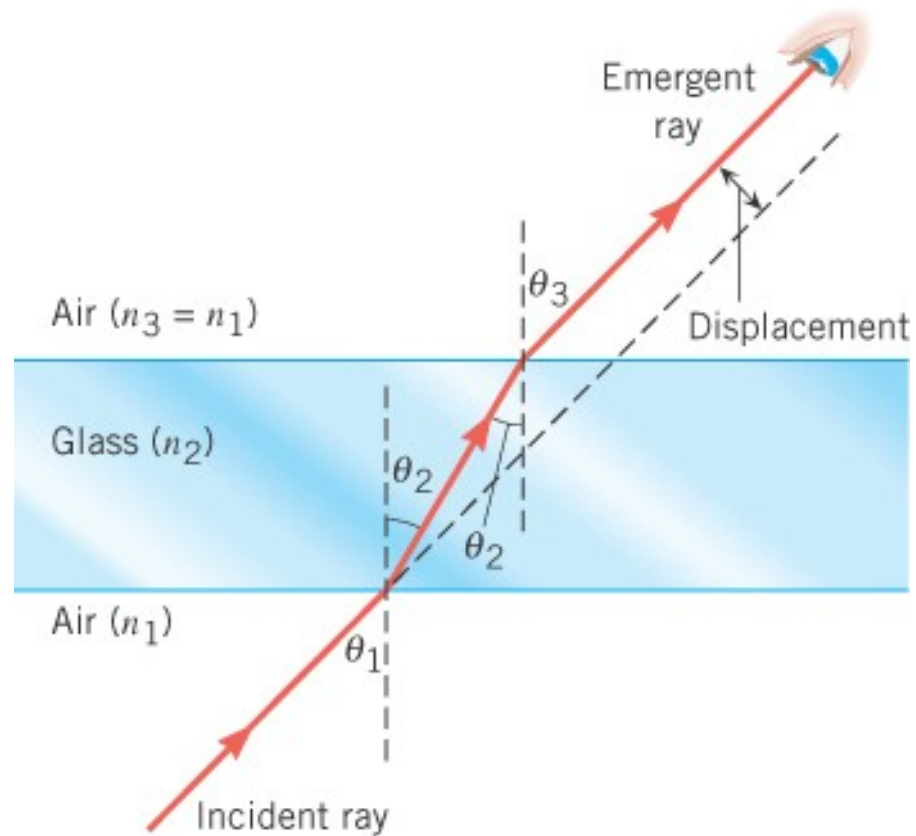
$$(a) \quad \sin \theta_2 = \frac{n_1 \sin \theta_1}{n_2} = \frac{(1.00) \sin 46^\circ}{1.33} = 0.54$$

$$\theta_2 = 33^\circ$$

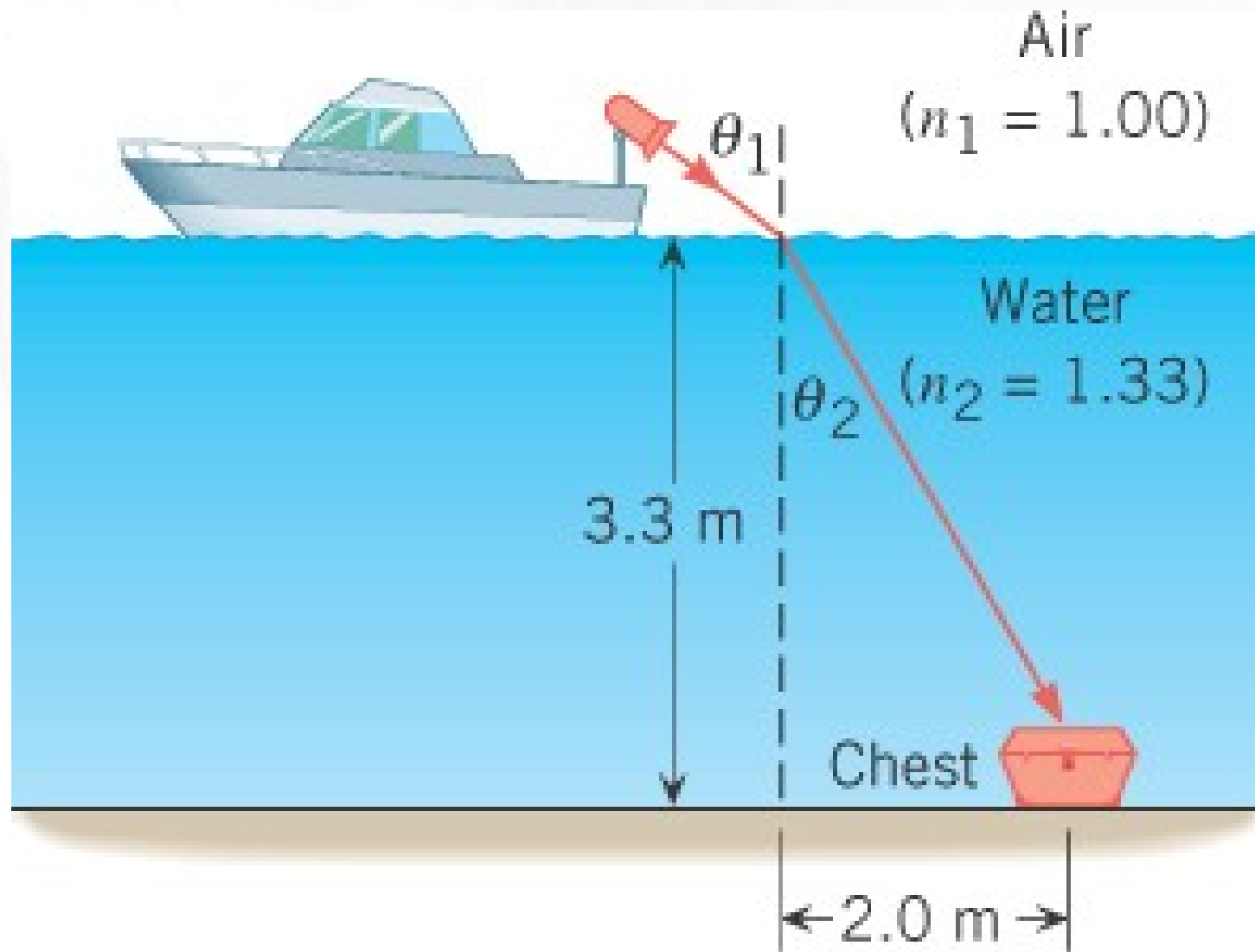
$$(b) \quad \sin \theta_2 = \frac{n_1 \sin \theta_1}{n_2} = \frac{(1.33) \sin 46^\circ}{1.00} = 0.96$$

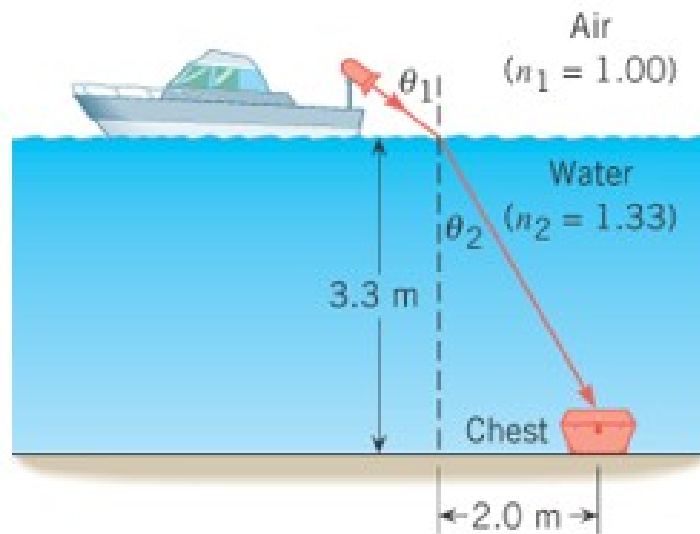
$$\theta_2 = 74^\circ$$

THE DISPLACEMENT OF LIGHT BY A SLAB OF MATERIAL



26.2 Snell's Law and the Refraction of Light





$$\theta_2 = \tan^{-1}(2.0/3.3) = 31^\circ$$

$$\sin \theta_1 = \frac{n_2 \sin \theta_2}{n_1} = \frac{(1.33) \sin 31^\circ}{1.00} = 0.69$$

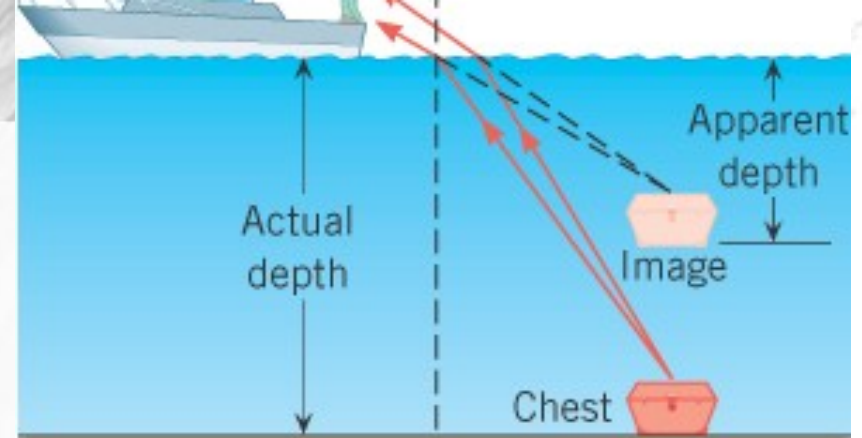
$$\theta_1 = 44^\circ$$

- ***This part is now optional***

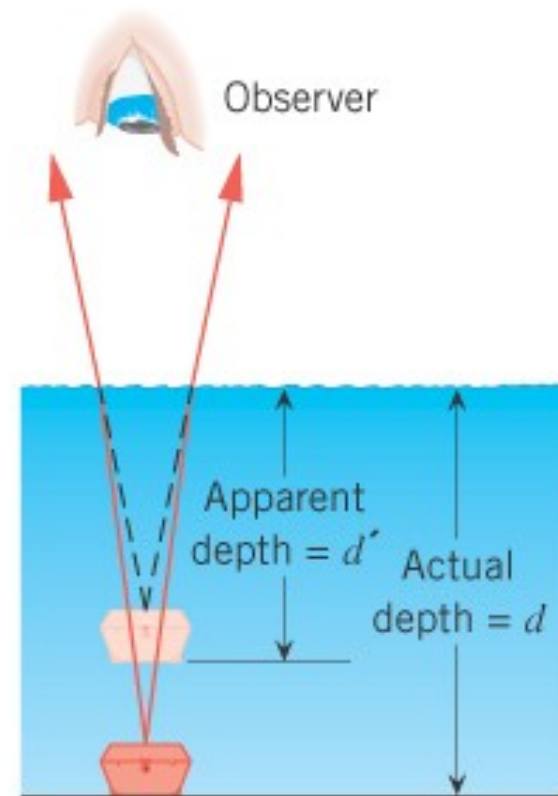
***Apparent depth,
observer directly
above object***

$$d' = d \left(\frac{n_2}{n_1} \right)$$

Draw the image of the chest



(a)



Example 3 The Apparent Depth of a Swimming Pool

A swimmer is treading water (with her head above the water) at the surface of a pool 3.00 m deep. She sees a coin on the bottom directly below. How deep does the coin appear to be?

Reasoning Equation 26.3 may be used to find the apparent depth, provided we remember that the light rays travel from the coin to the swimmer. Therefore, the incident ray is coming from the coin under the water ($n_1 = 1.33$), while the refracted ray is in the air ($n_2 = 1.00$).

Example 3 The Apparent Depth of a Swimming Pool

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Solution The apparent depth d' of the coin is

$$d' = d \left(\frac{n_2}{n_1} \right) = (3.00 \text{ m}) \left(\frac{1.00}{1.33} \right) = \boxed{2.26 \text{ m}} \quad (26.3)$$

Conceptual Example 4 On the Inside Looking Out

A swimmer is under water and looking up at the surface. Someone holds a coin in the air, directly above the swimmer's eyes. To the swimmer, the coin appears to be at a certain height above the water. Is the apparent height of the coin greater, less than, or the same as its actual height? **DRAW**



1) A fish lies 60cm below the surface of a clear pond. If the index of refraction of water is assumed to be 1.33 and that of air is approximately 1.

A) A person is looking at the fish from above. (right above). Draw the 2 rays coming from the fish to the eye and draw the image of the fish.

$$d' = d \left(\frac{n_2}{n_1} \right)$$

B) how far below the surface does the fish appear to a person looking down from above.
!!! The light travels from WATER (1.33) to AIR (1) !!! THINK !!!

2) A rock appears to lie just 24cm below the surface of a smooth stream when viewed from above the surface of the stream. Using the indices of refraction given in 1) What is the actual distance of the rock below the surface?

3) An insect is embedded inside a glass block (1.5) so that it is located 2.4 cm below a plane surface of the block. How far from the surface does this insect appear to a person looking at the block. DRAW.

4) A man with a height of 1.8m stands 3m in front of a plane mirror viewing his image. How tall is the image, and how far from the man is the image located?(see next slide)

5) A light ray enters a piece of crown glass at an angle of 50 and is refracted to 30 degrees inside the glass. What is the index of refraction of the glass? DRAW.

6) If a ray enters the glass in 5) at an angle of 20 degrees, what will be the angle of refraction be ? DRAW

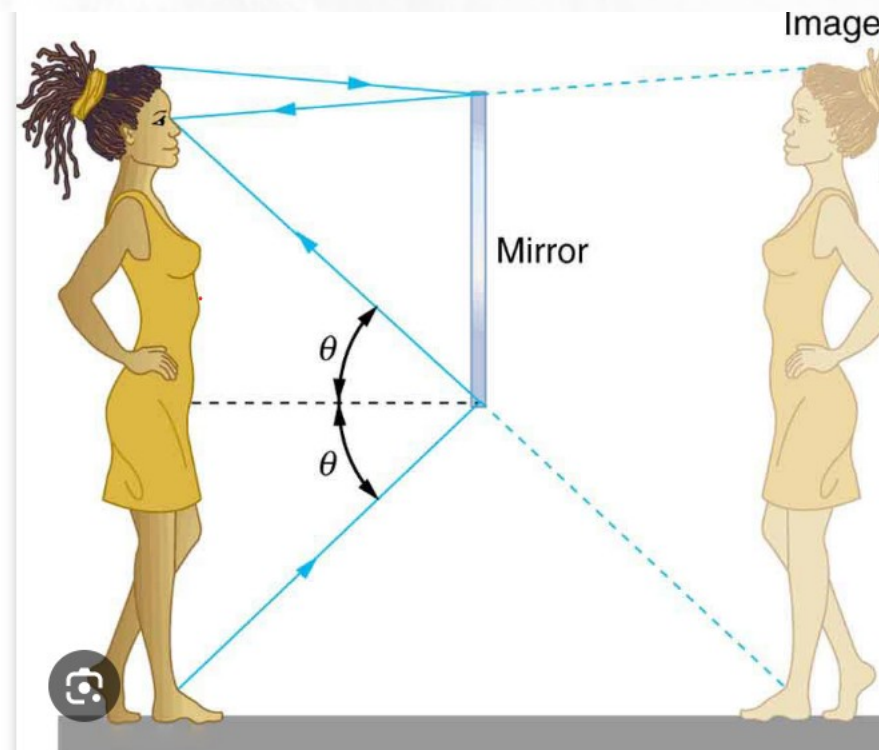
7) The index of refraction of water is 1.33. What is the speed of light in water ?

8) A ray of light passes from water (1.33) into a sheet of flint glass ($n=1.61$). Making an angle of incidence of 65. What is the angle of refraction (DRAW)

9) A block of crown glass ($n=1.63$) is immersed in an unknown liquid. A ray of light is measured to make an angle of incidence of 48 degrees as it approaches the glass. The angle in the glass is 36. What is the index of refraction of the liquid ?

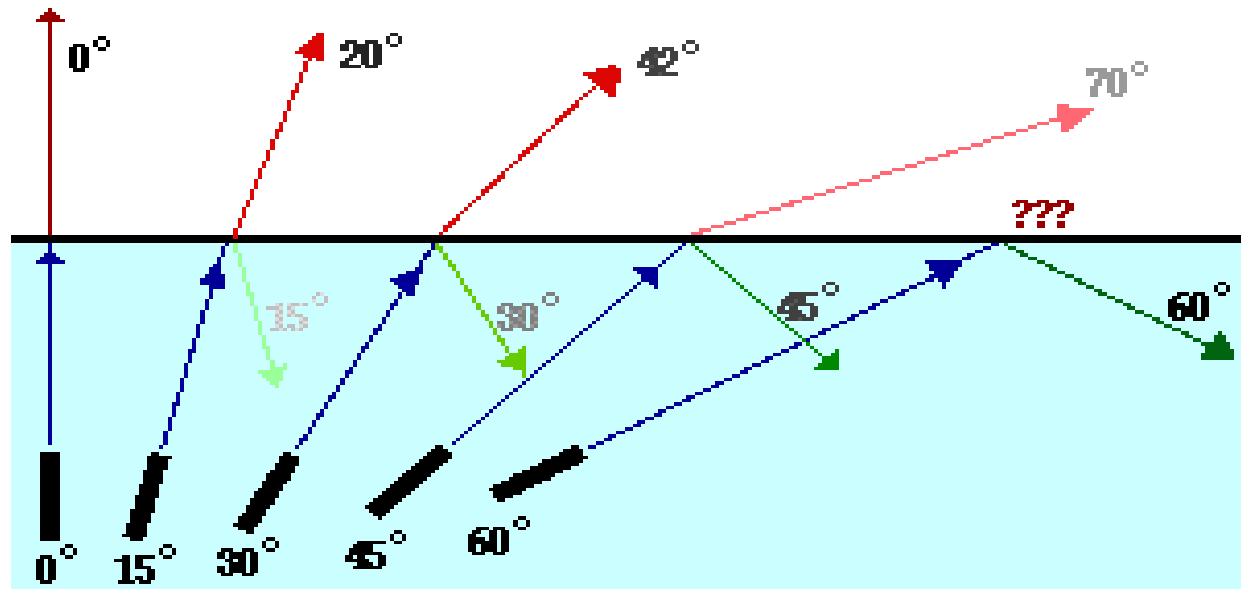
10) A diamond ($n=2.42$) is in water ($n=1.33$), and a ray of light shines on it, making an angle of incidence of 55. What is the angle of refraction inside the diamond? DRAW

Virtual image. Example: plane mirror.



Total Internal Reflection

When light passes from a medium of larger refractive index into one of smaller refractive index, the refracted ray bends away from the normal. **As the angle of incidence increases from 0 to greater angles ...**



**...the refracted ray becomes dimmer (there is less refraction)
...the reflected ray becomes brighter (there is more reflection)
...the angle of refraction approaches 90 degrees until finally
a refracted ray can no longer be seen.**

https://phet.colorado.edu/sims/html/bending-light/latest/bending-light_en.html

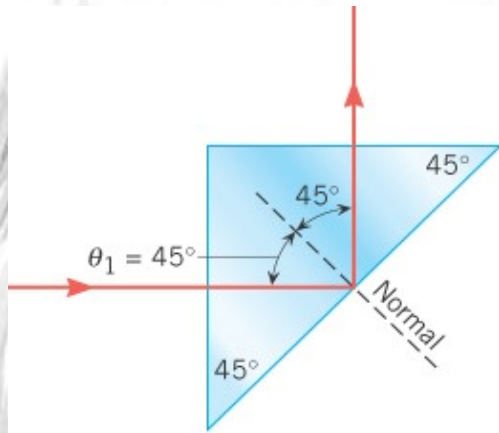
Use the applet 1) to show this

Phenomenon 2) use the sensor

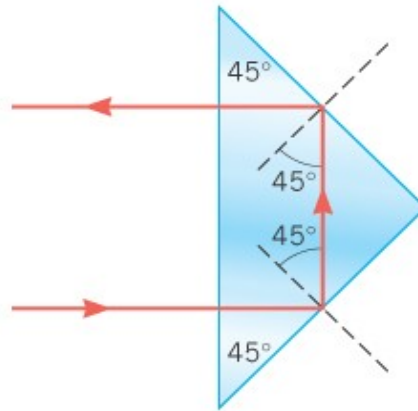
To show the reflected ray becomes brighter.

3) show how a prism can be used as a mirror to reflect light. This is used in binocular !

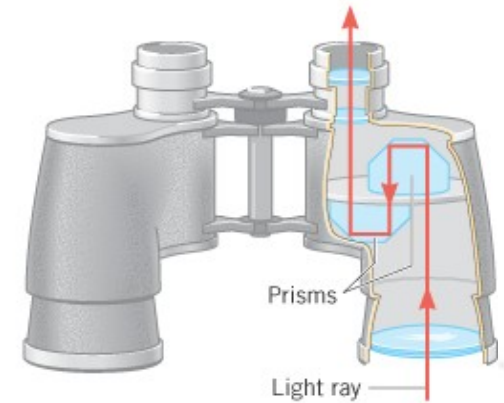
Total Internal Reflection : APPLICATION – do demo



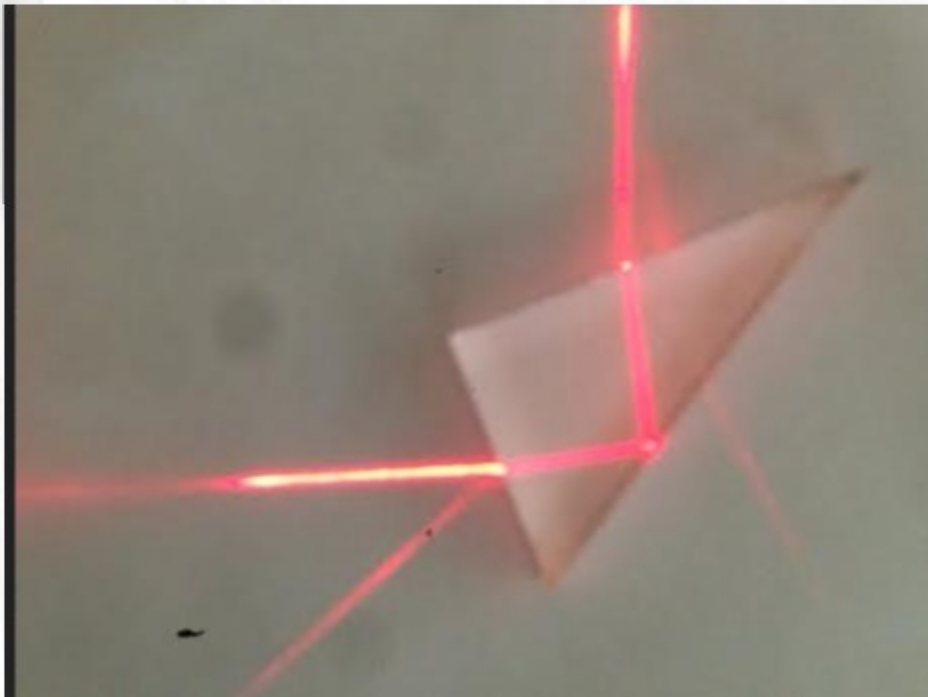
(a)



(b)



(c)



***We can use a prism 45-45-90
As a totally reflecting surface.***

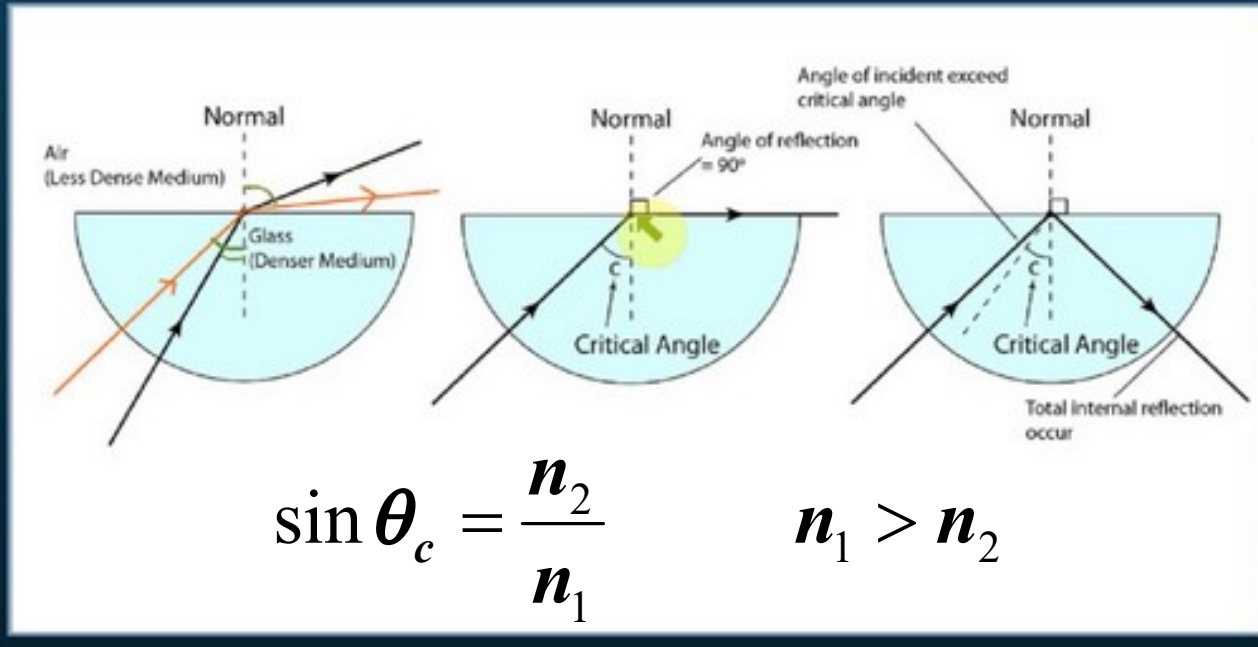
Its called a porro prism.



TOTAL INTERNAL REFLECTION

Critical Angle

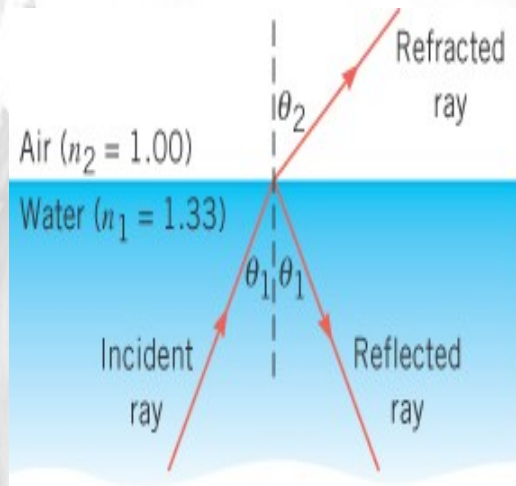
The critical angle is the angle of incident in an optically denser medium for which the angle of refraction is 90° .



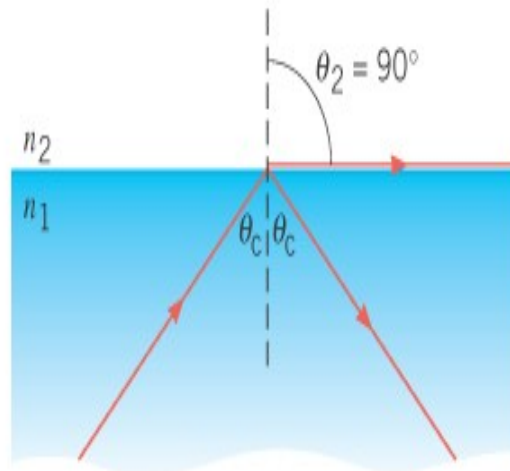
If incidence angle $<$ critical angle $\rightarrow$ it is refracted and reflected

If incidence angle $=$ critical angle $\rightarrow$ it is refracted at 90° and reflected

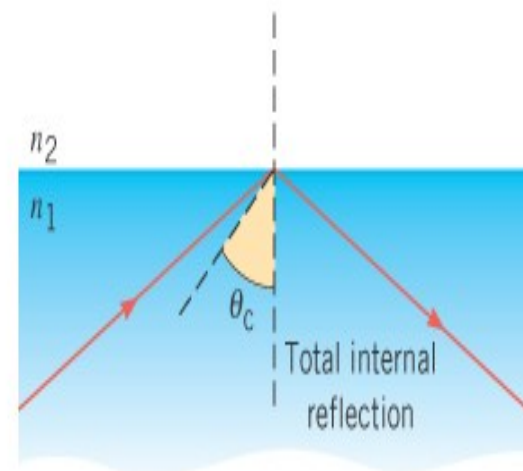
If incidence angle $>$ critical angle $\rightarrow$ all is reflected



(a)



(b)



(c)

Total Internal Reflection

Snell's law: $n_1 \sin \theta_1 = n_2 \sin \theta_2$

At critical angle: $\theta_2 = 90^\circ \rightarrow \sin \theta_2 = 1$

So: $n_1 \sin \theta_c = n_2 \rightarrow \sin \theta_c = n_2 / n_1$

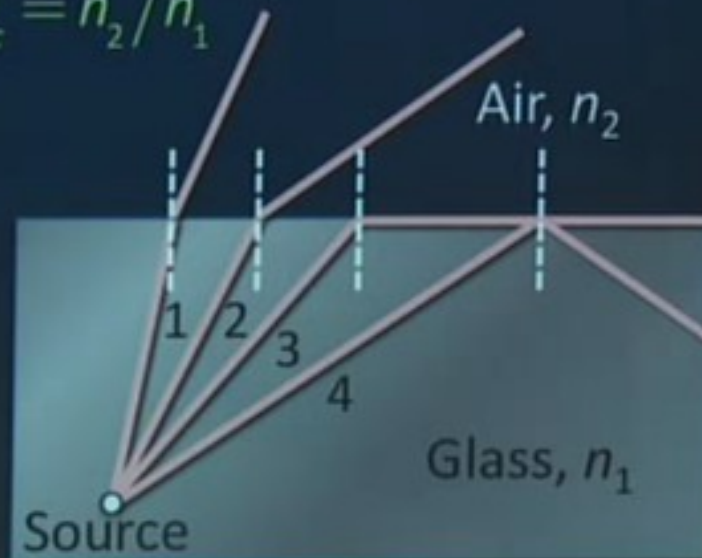
Do the numbers:

$$n_1 = n_{\text{glass}} \approx 1.5$$

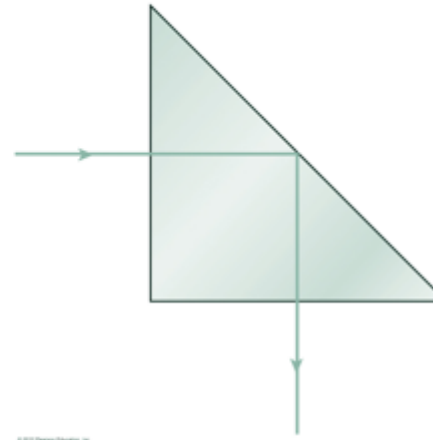
$$n_2 = n_{\text{air}} \approx 1$$

$$\sin \theta_c = 1/1.5 = 0.67$$

$$\theta_c = \sin^{-1}(0.67) = 42^\circ$$

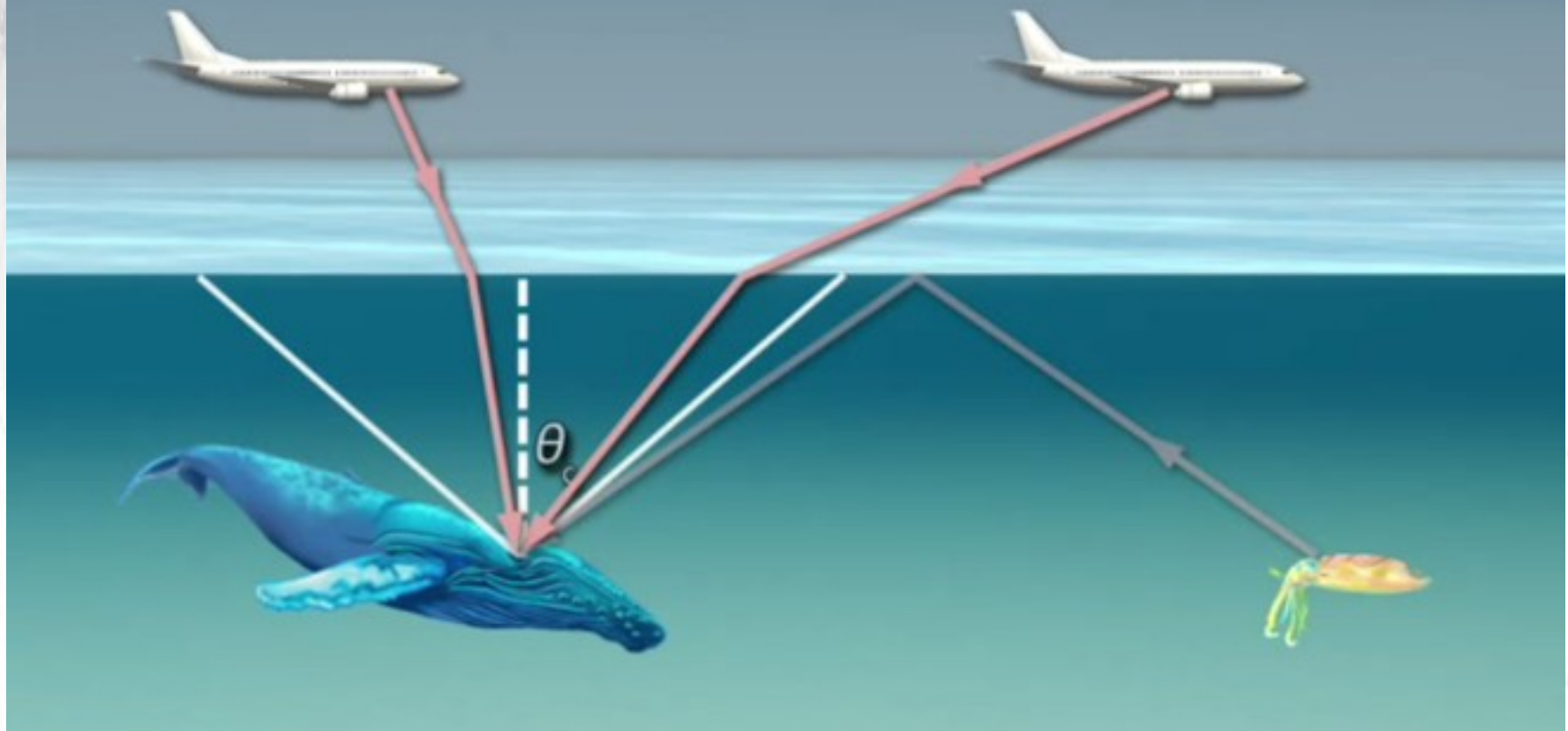


- The glass prism in the figure has $n = 1.5$ and is surrounded by air ($n = 1$). What would happen to the incident light ray if the prism were immersed in water ($n = 1.333$)?
- A. Most of it would emerge into the water through the diagonal face and some would be reflected as shown.
- B. Most of it would be reflected as shown and some would emerge into the water through the diagonal face.



The critical angle for water is measured between the beam and a line perpendicular to the surface, and is 49 degrees

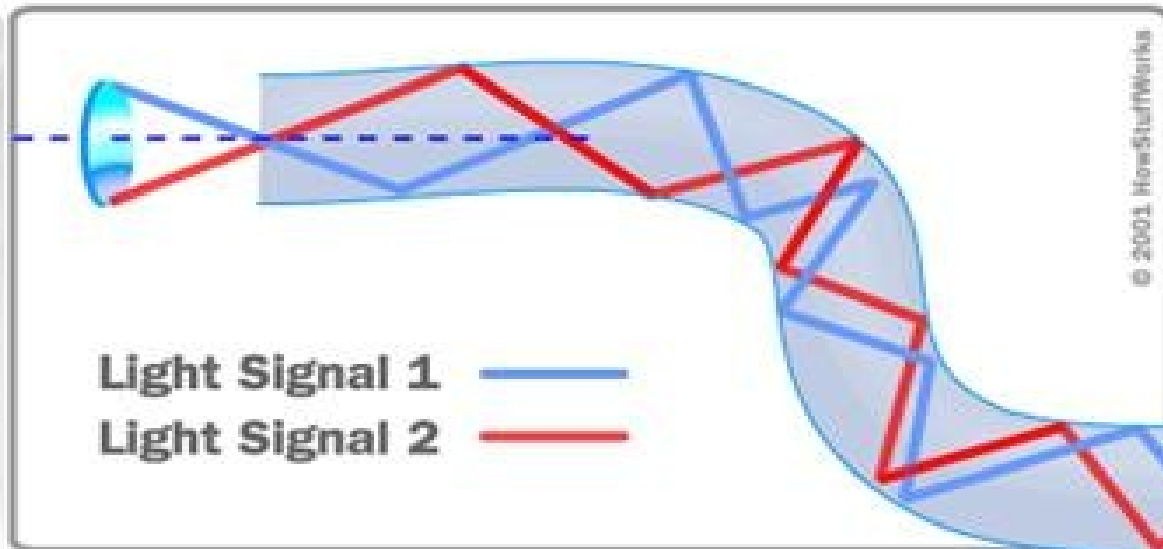
Whale Watch!



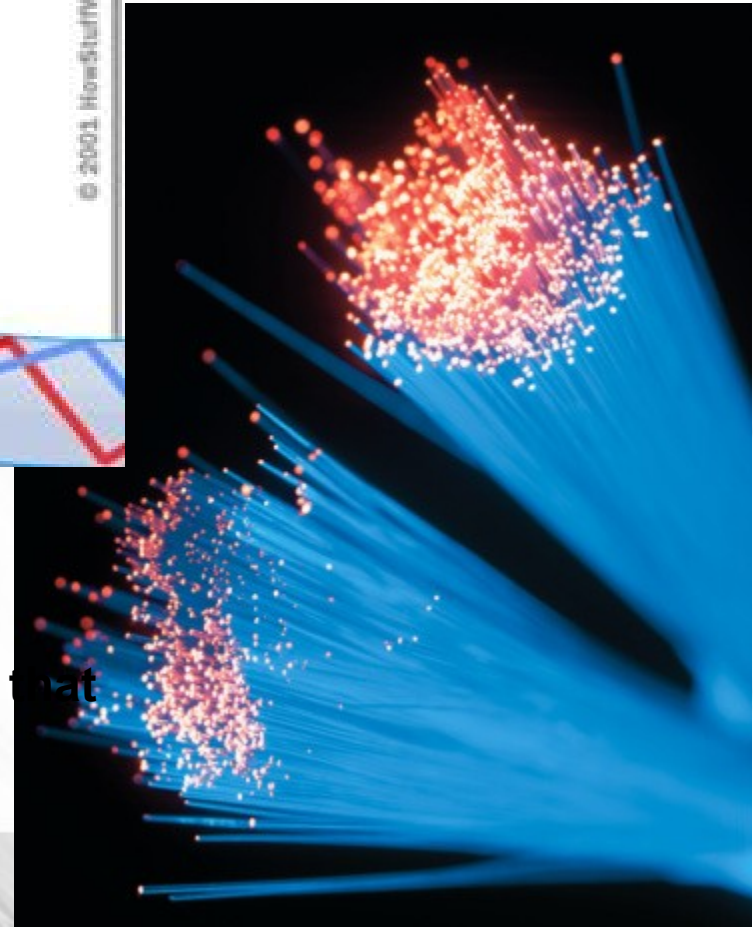
TOTAL internal reflection

Application with light waves.

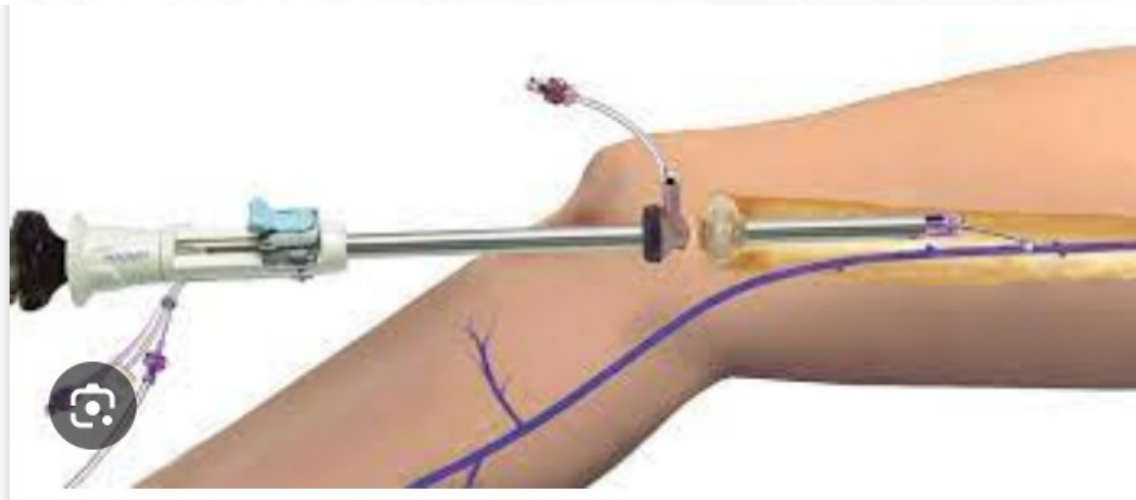
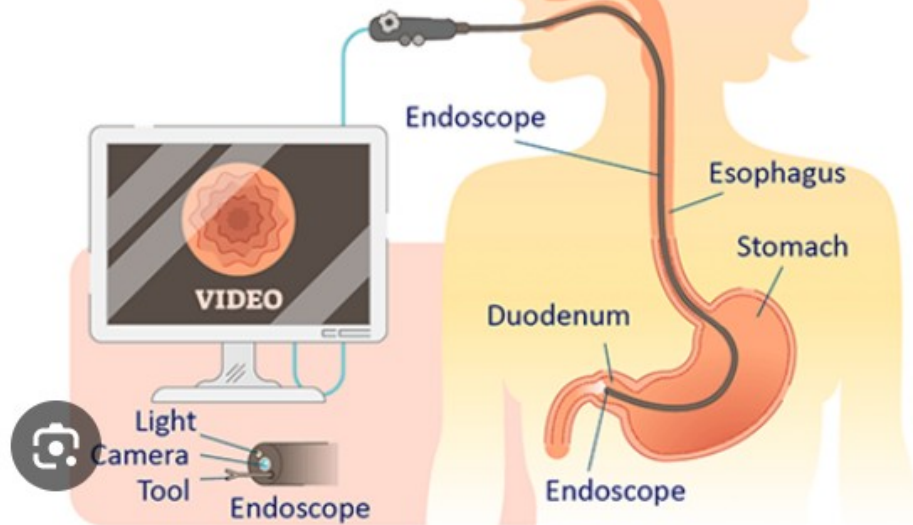
You can trap light in optic fibers using the bending of light. In that case you get a total reflection and the light is trapped. Information can travel a long distance without loss of energy.



In some conditions, there is so much bending that the waves are reflected. Here: light.

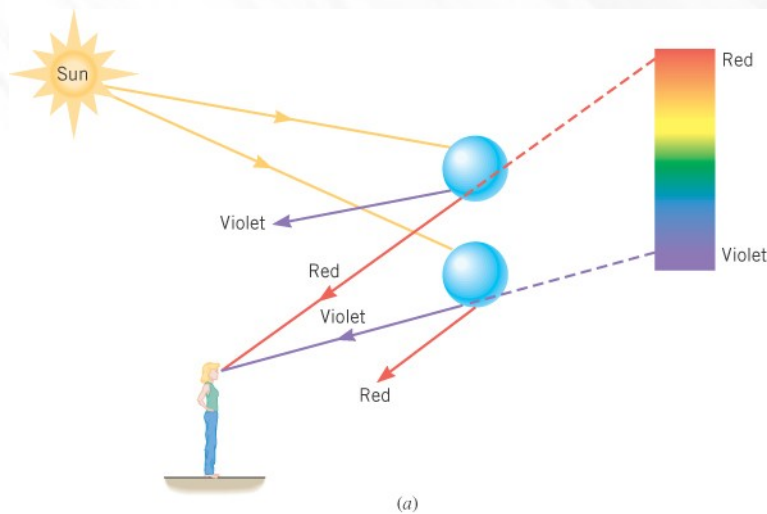
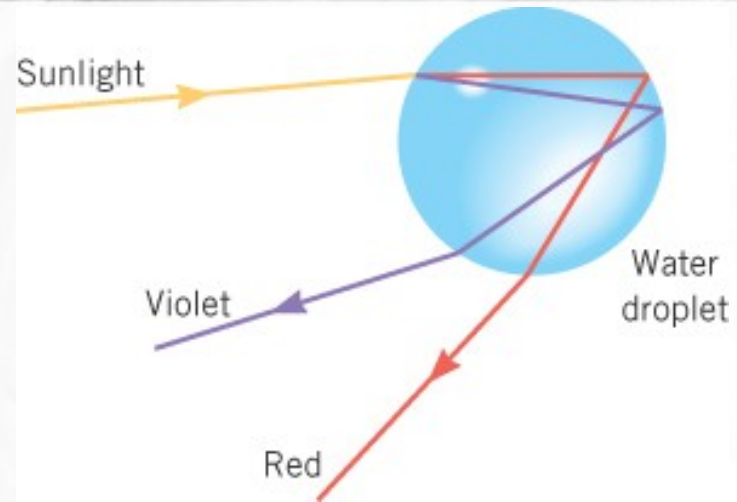
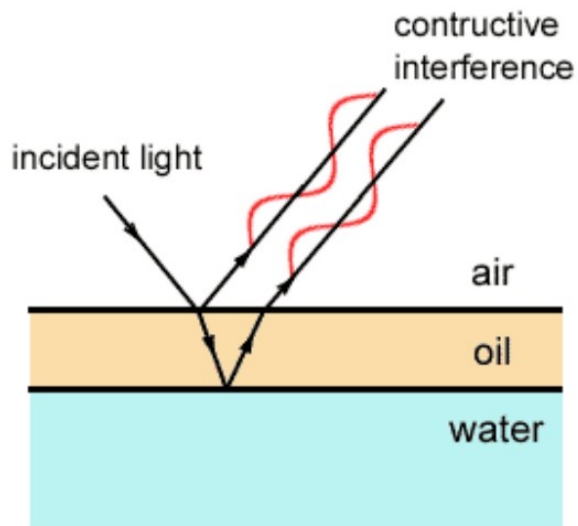


ENDOSCOPY



26.5 The Dispersion of Light: Prisms and Rainbows

See exploration physical



TOTAL internal reflection
Application with sound waves.
SOUND CHANNEL

Sound can also be trapped in a channel called the sound channel.
This sound channel is used by whales to communicate from
Very far apart. The sound channel is also used by submarine to hide.
Their noise go undetectable.

Sound travels travel fast = water warmer



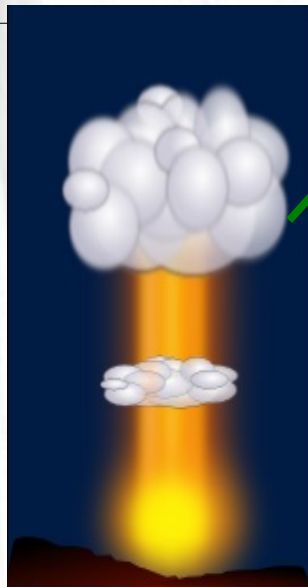
Sound travels travel slow = water cooler

The yellow submarine goes undetected.

Sound travels travel fast = more pressure

**In some conditions, there is so much bending that
The waves are reflected. Here: sound**

Same principle works in the atmosphere. Maurice Ewing used this principle
To spy on the nuclear tests performed by the soviet union during the cold war
. He used A giant microphone attached to an air balloon and placed in the
Sound channel of the atmosphere.



Warmer air = ozone layer

cooler air

Warmer air

One day on of the disk crashed on the ground
Near Roswell. People thought it was an UFO .
Military didn't dismiss the idea because the mission was
Secret. Then the whole story about Roswell's alien



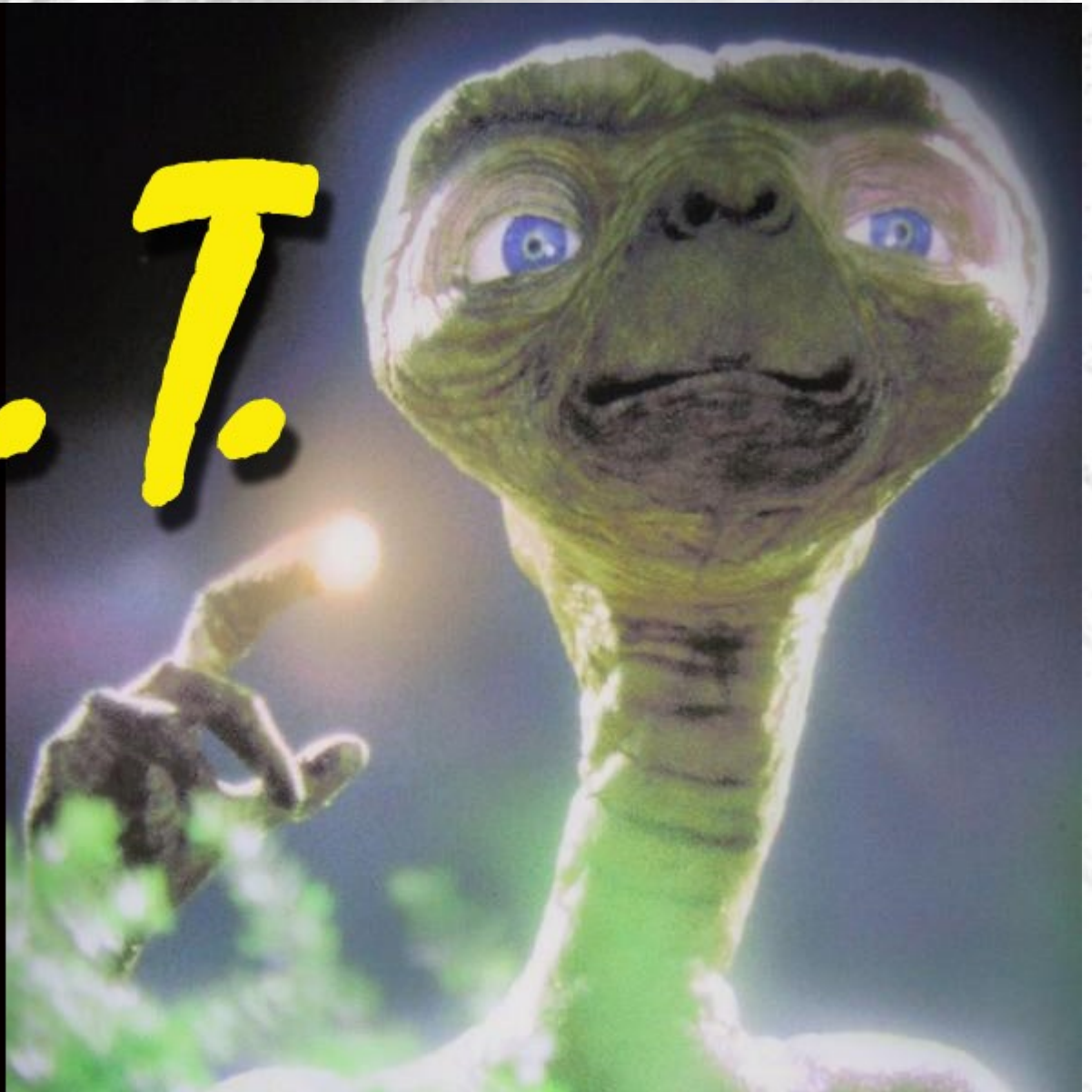
They did it !



**USSR home of
Stalin**

**Secret base =
Roswell, New Mexico**

E.T.



white light

ROYGBIV



Dispersion for diamond is
Also large → rainbow.

A diamond has a critical angle = 25 degrees.
The colors are separated
And keep reflecting inside the
Diamond. It gives the diamond this
Shiny look. But today the zirconium
is much better.

DeBeers monopoly wants people
Think otherwise.

40

Zr

Zirconium

91.22

2-8-18-10-2



Watch *blood diamonds* and
Buy zirconium !!

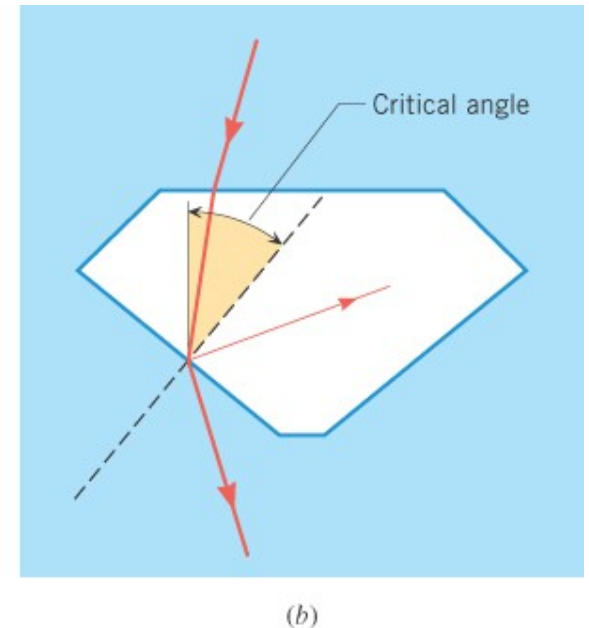
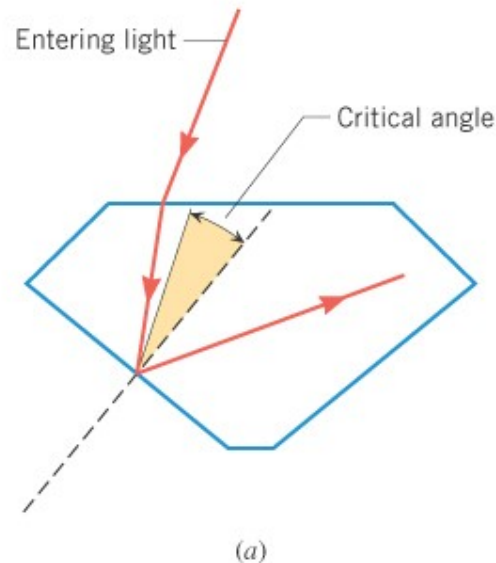


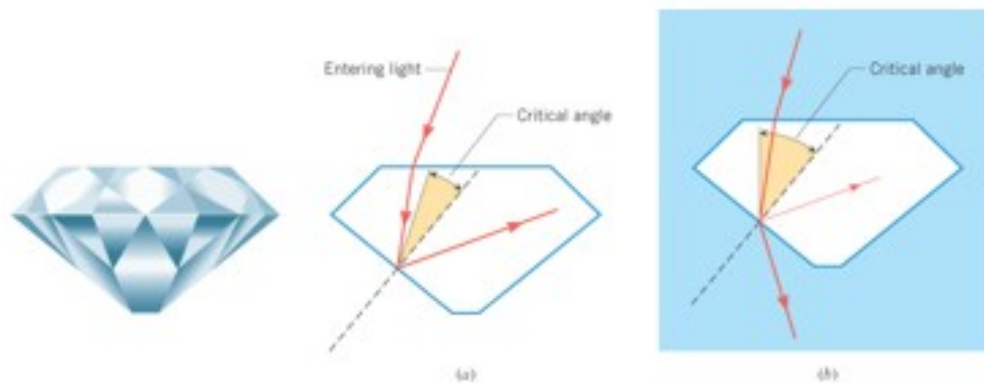
[http://en.wikipedia.org/wiki/
Blood_diamond](http://en.wikipedia.org/wiki/Blood_diamond)

26.3 Total Internal Reflection

Example 5 Total Internal Reflection

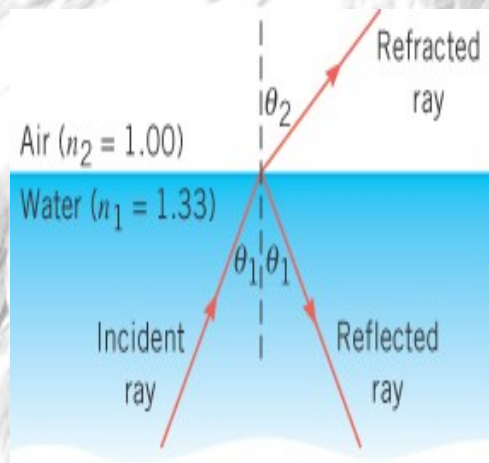
A beam of light is propagating through diamond and strikes the diamond-air interface at an angle of incidence of 28 degrees. (a) Will part of the beam enter the air or will there be total internal reflection? (b) Repeat part (a) assuming that the diamond is surrounded by water.



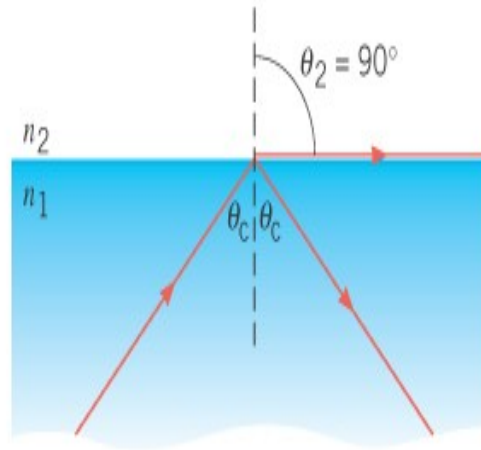


(a)
$$\theta_c = \sin^{-1}\left(\frac{n_2}{n_1}\right) = \sin^{-1}\left(\frac{1.00}{2.42}\right) = 24.4^\circ$$

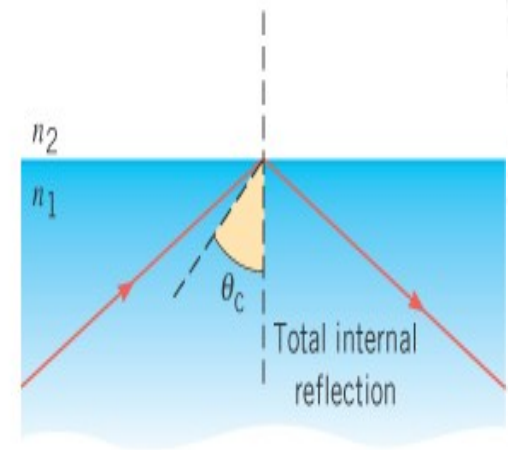
(b)
$$\theta_c = \sin^{-1}\left(\frac{n_2}{n_1}\right) = \sin^{-1}\left(\frac{1.33}{2.42}\right) = 33.3^\circ$$



(a)



(b)



(c)

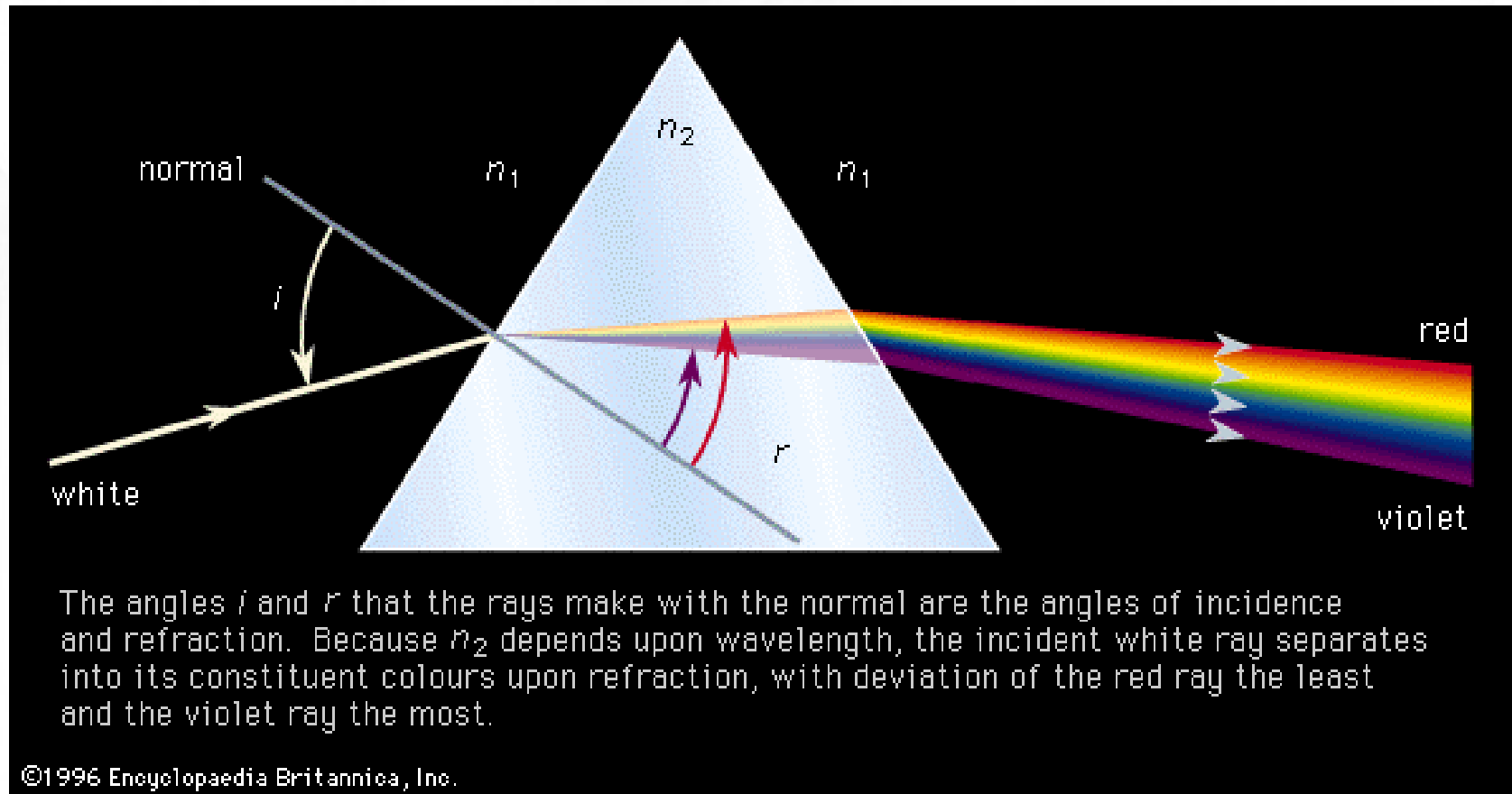
1) see the drawing above. The light travels from water (1.33) to air (1). As you increase the angle of incidence, the Angle of refraction increases. When the refraction angle is 90 degrees, the light is reflected and trapped. The light can't reach air. A) Using Snell's law with refraction angle = 90 find the critical angle of water. (from 1.33 to 1)
 B) that means if the angle of refraction is larger than the critical angle, the light is totally reflected.
 If the angle is 30, is the light refracted ? What if the angle is 60 ?

2) What is the critical angle of incidence fro crown glass ($n=1.63$) ?

3) What index of refraction must a transparent material have in order for the total internal reflection to take place at all angles larger then 50 degrees ?

4) what is the critical angle of incidence of diamond ($n=2.42$) ?
 What do diamond sparkles ? But so zirconium and it is cheaper !

**Refraction of light explains the rainbows and spectra.
Newton was the one to understand this process.**



How much bending light undergoes depends on the wavelength of the colors that make up white light. Blue bends more than red. The speed of light is smaller in glass than in air. So refraction takes place. This is how a rainbow is produced.

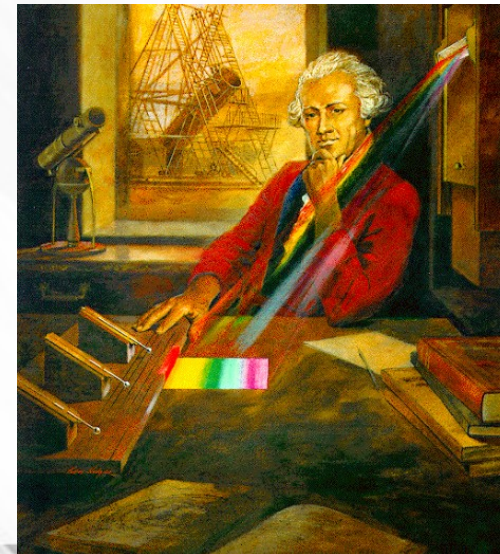


Newton found out that white light is composed of the color of the rainbow. Each color will be bent by glass or drops of water by a different amount.



Christiaan Huygens and Newton disagree
On the nature of light.
Huygens supported the wave
Nature of light. Newton
The particle nature of light
That he called corpuscles
First, Newton's view was accepted
But rejected after experiments
Done by Young. Young showed that
Light behaves like waves.
(light can interfere)

1905 Einstein describe light as units of
Energy or bundle. These units are called
Photons or wave-particles = wave packet.



<https://www.britannica.com/science/light/Youngs-double-slit-experiment>

26.5 The Dispersion of Light: Prisms and Rainbows

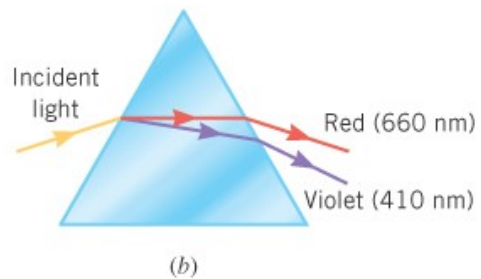
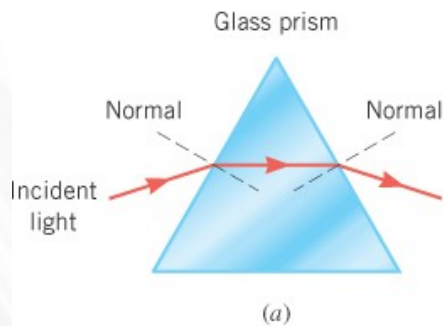


Table 26.2 Indices of Refraction n of Crown Glass at Various Wavelengths

Approximate Color	Wavelength in Vacuum (nm)	Index of Refraction, n
Red	660	1.520
Orange	610	1.522
Yellow	580	1.523
Green	550	1.526
Blue	470	1.531
Violet	410	1.538

Lenses / FORMATION of IMAGES

Lenses refract light in such a way that an image of the light source is formed.

With a converging lens (convex), paraxial rays that are parallel to the principal axis converge to the focal point.

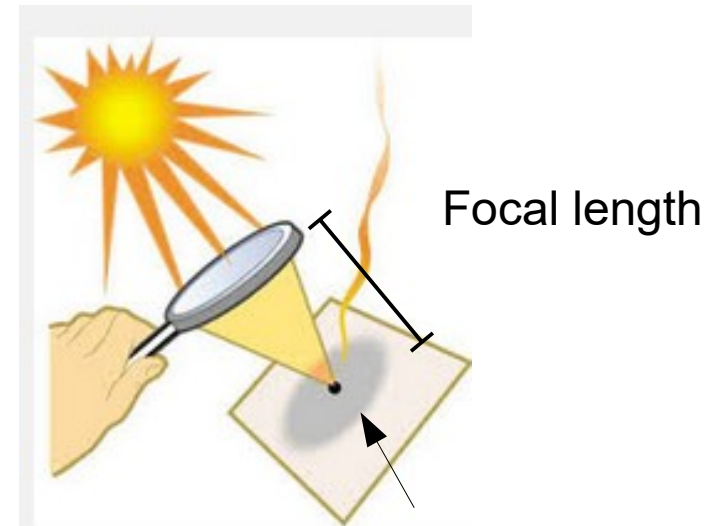
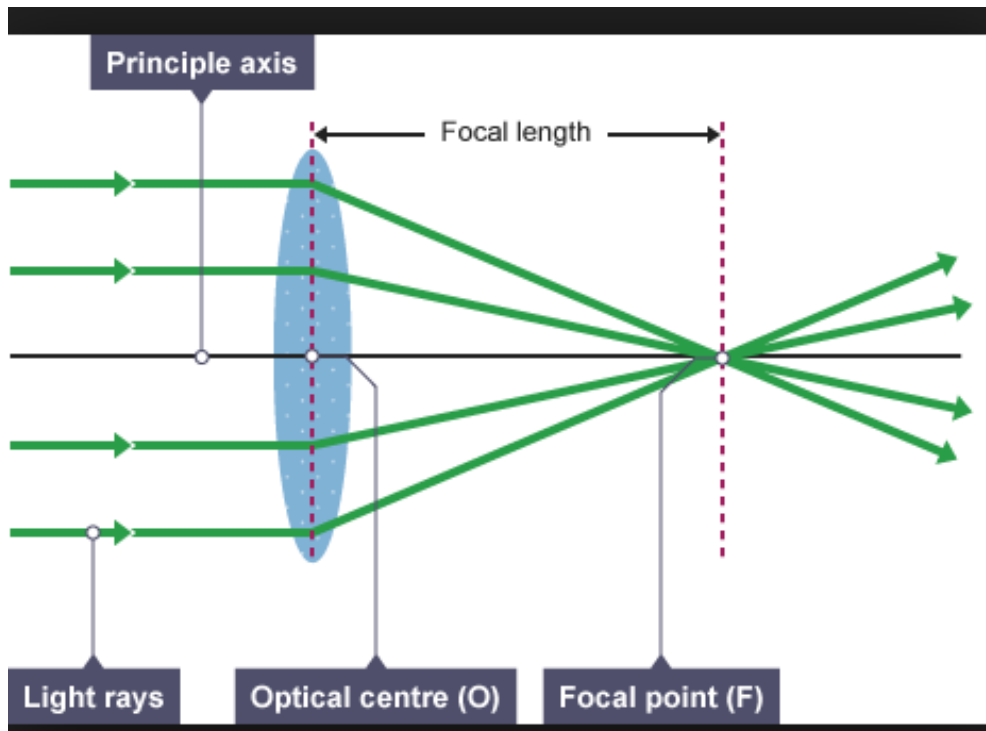
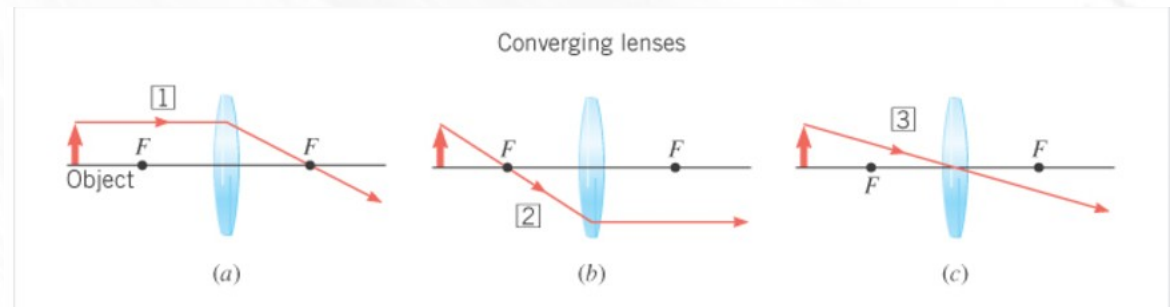
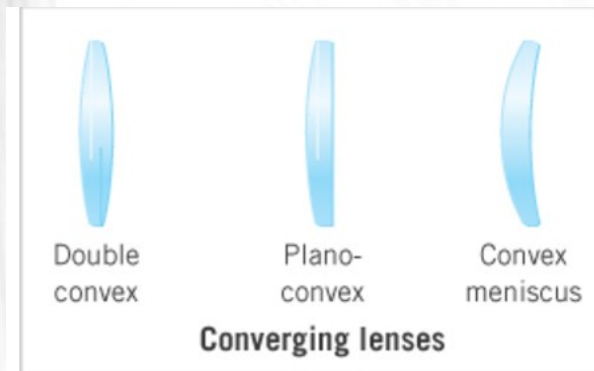
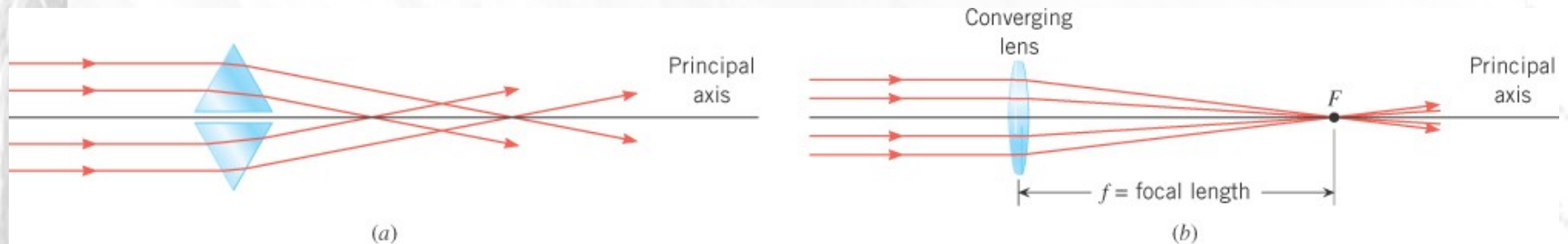


Image of the sun upside down

Do demo This is why you can burn a paper with a magnifying glass

Lenses refract light in such a way that an image of the light source is formed.

With a converging lens, paraxial rays that are parallel to the principal axis converge to the focal point.



Converging lenses

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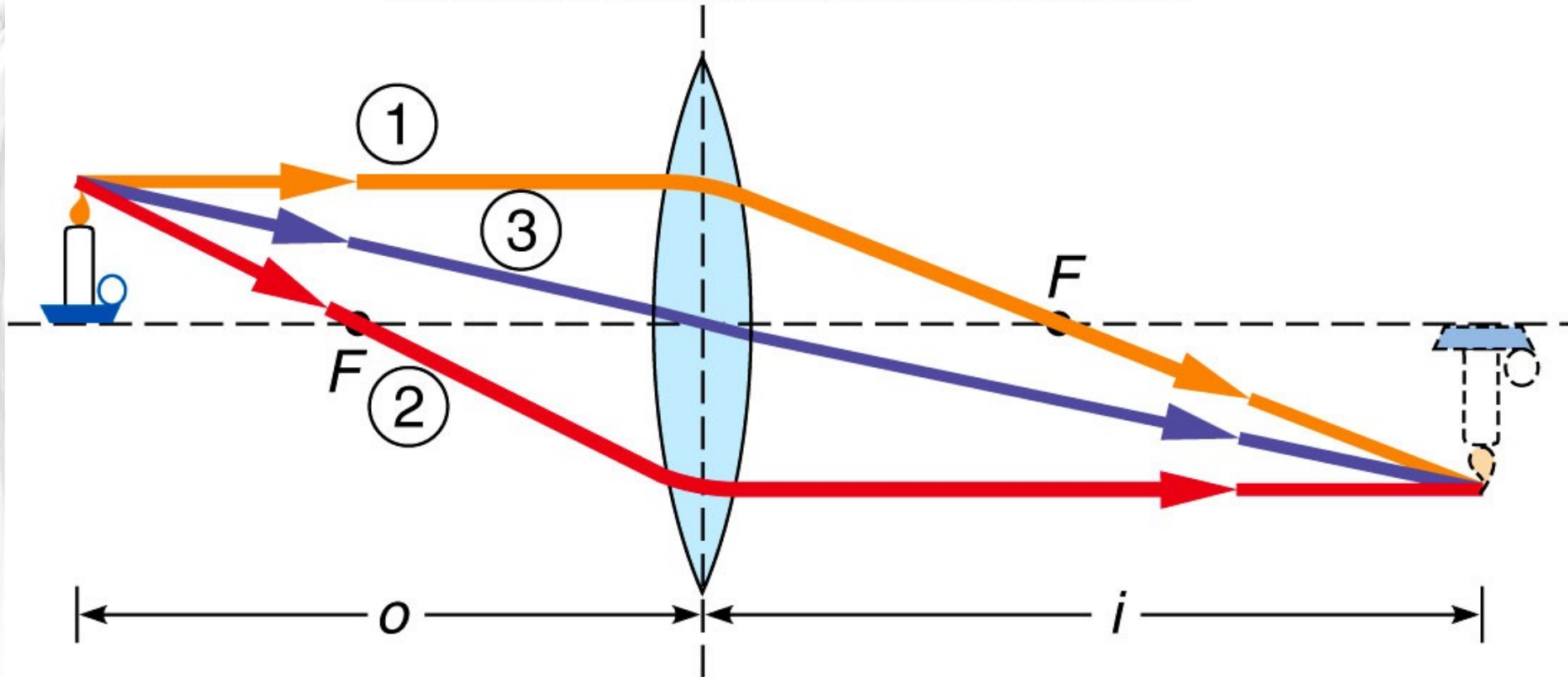
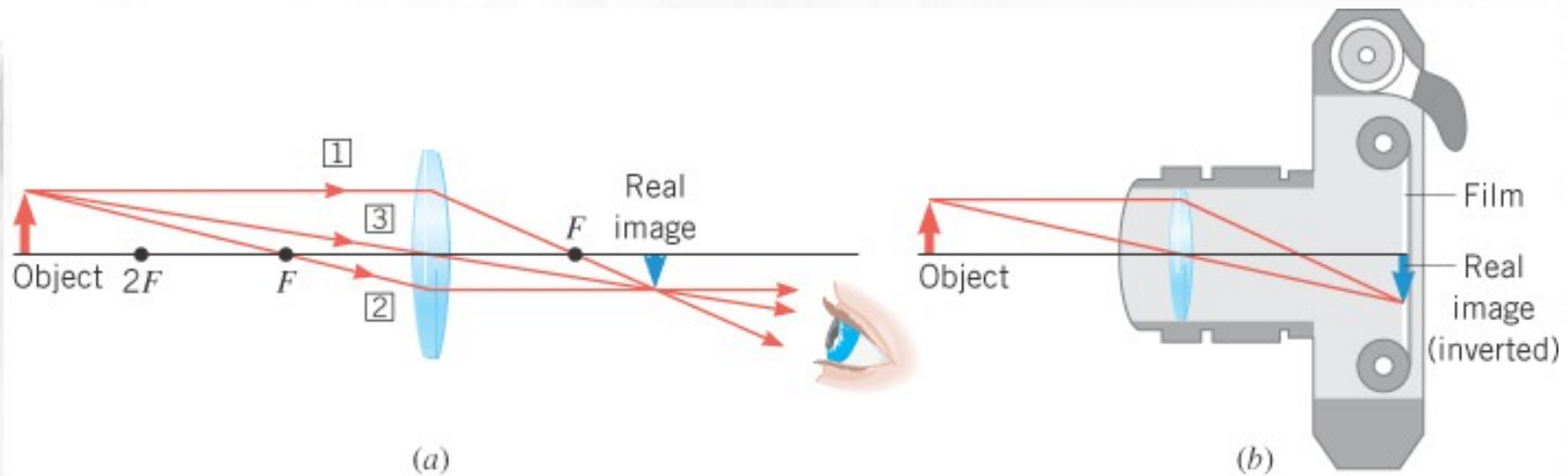
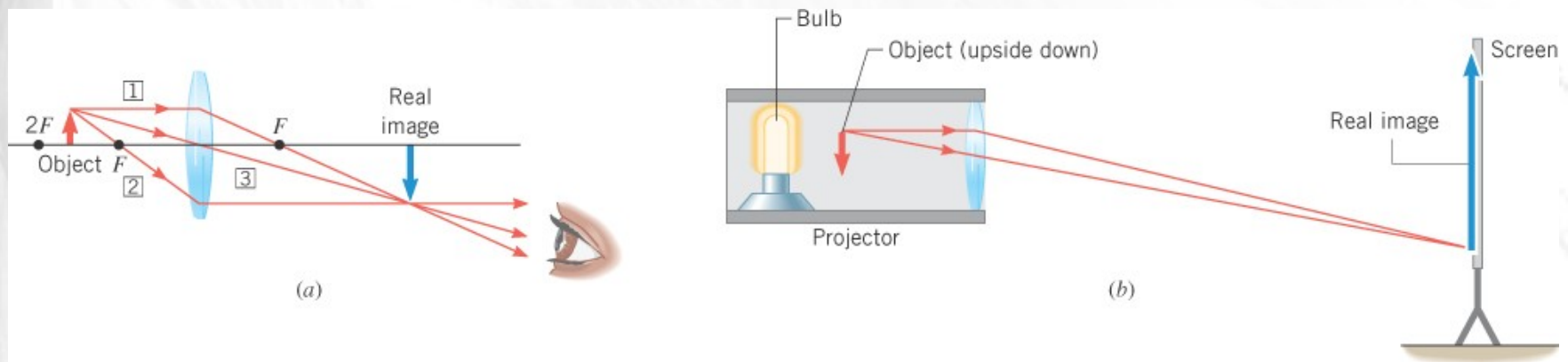


IMAGE FORMATION BY A CONVERGING LENS

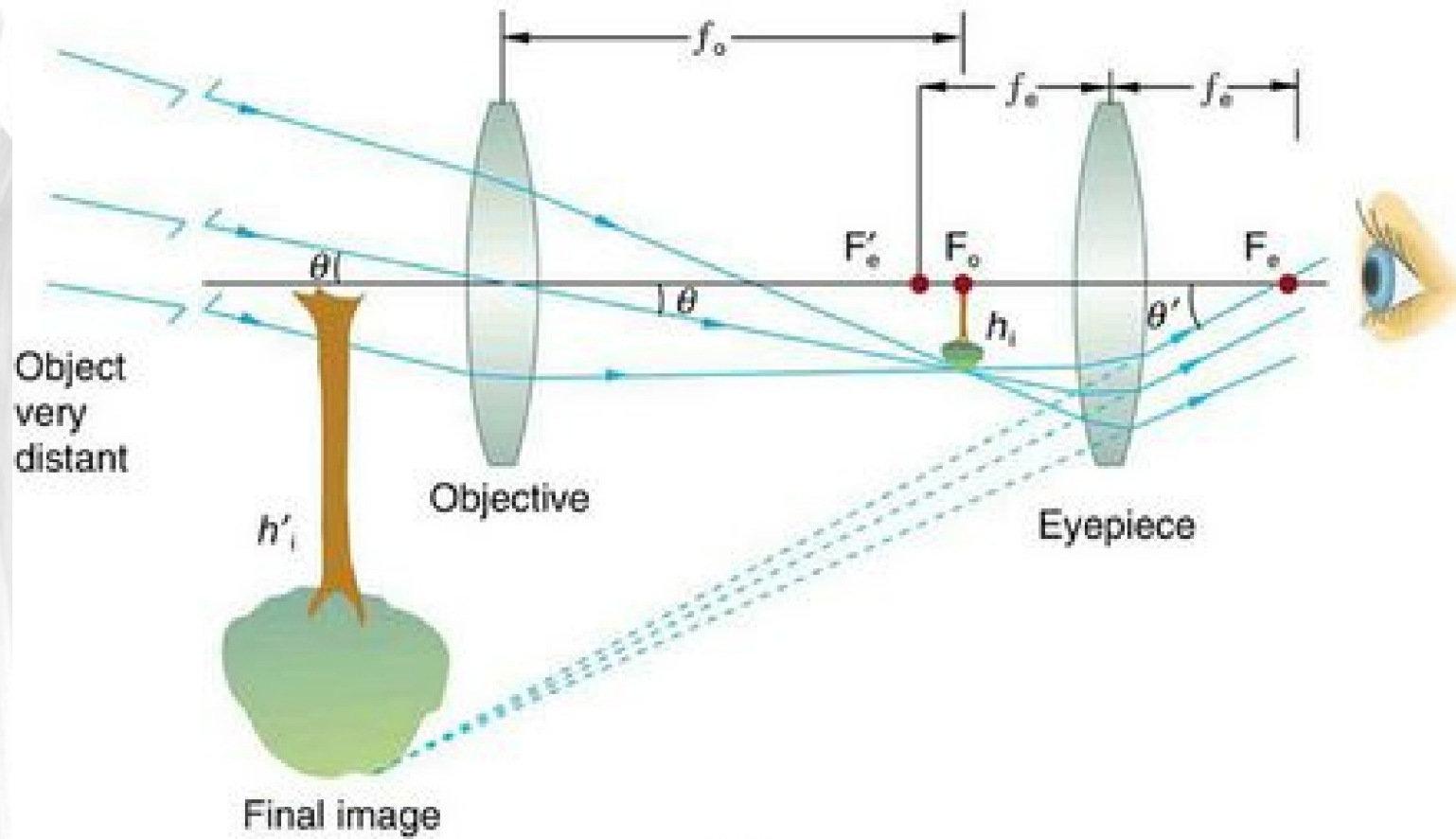


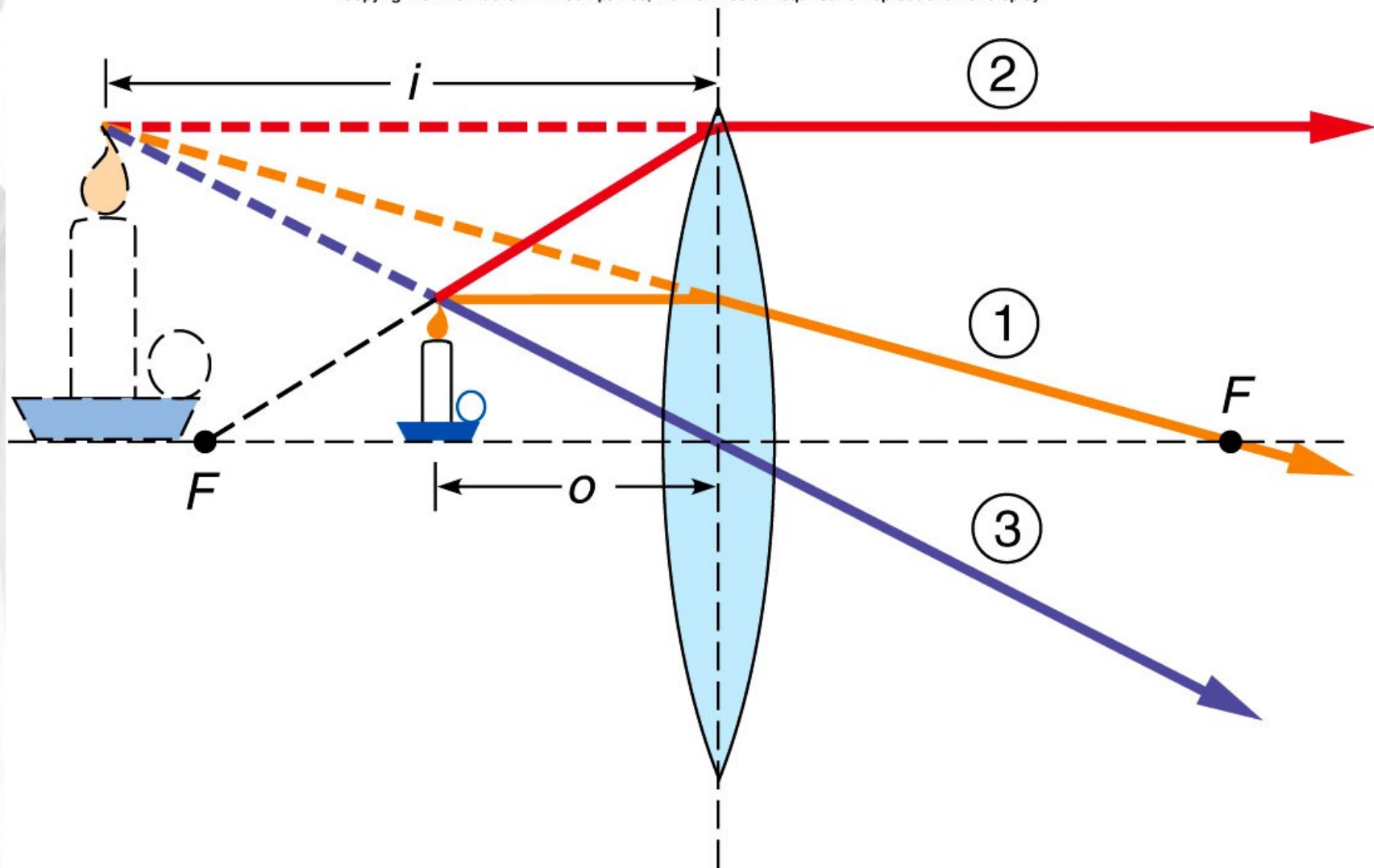
In this example, when the object is placed further than twice the focal length from the lens, the real image is inverted and smaller than the object.

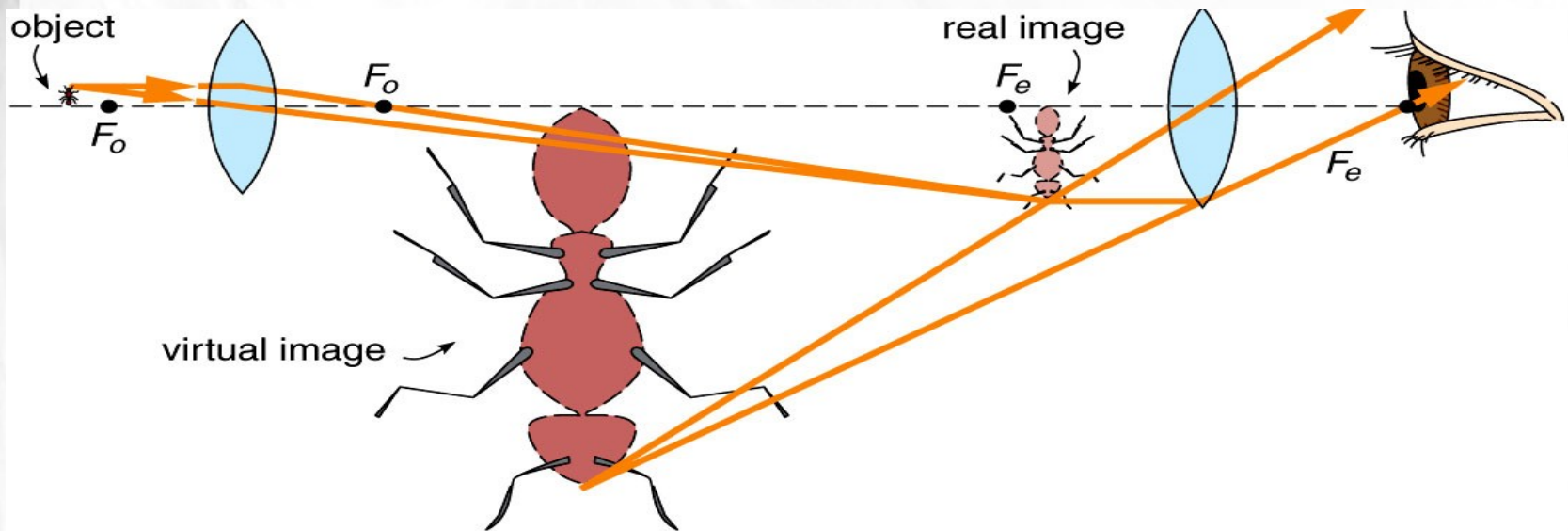
26.7 The Formation of Images by Lenses



When the object is placed between F and $2F$, the real image is inverted and larger than the object.



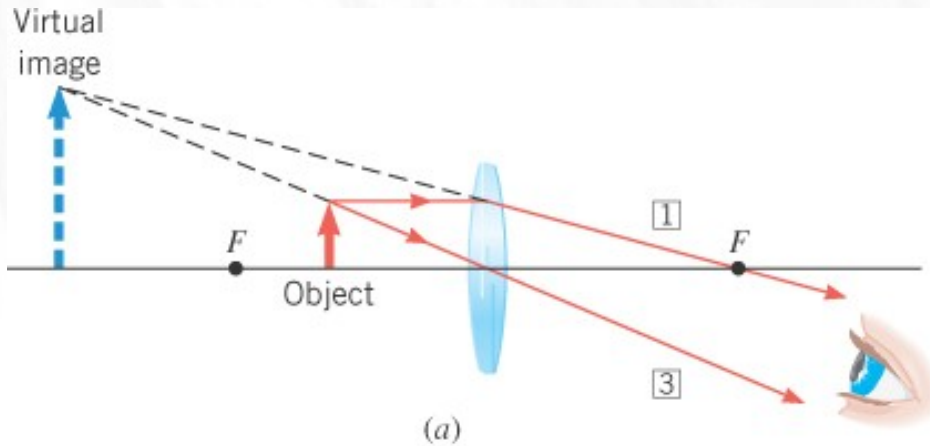




Ref: The physics of every day phenomena
Thomas Griffith

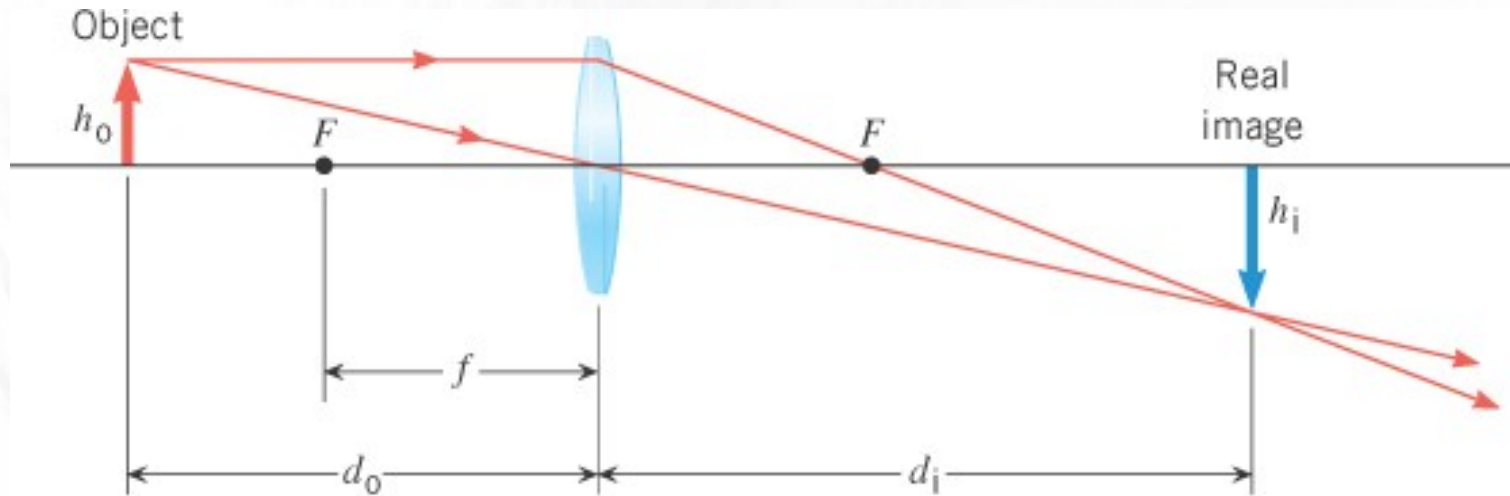
WHAT Are THESE OPTICAL INSTRUMENTS ?

26.7 The Formation of Images by Lenses



When the object is placed between F and the lens, the virtual image is upright and larger than the object.

26.8 The Thin-Lens Equation and the Magnification Equation



$$\frac{1}{d_o} + \frac{1}{d_i} = \frac{1}{f}$$

$$m = \frac{h_i}{h_o} = -\frac{d_i}{d_o}$$

***Example 9* The Real Image Formed by a Camera Lens**

A 1.70-m tall person is standing 2.50 m in front of a camera. The camera uses a converging lens whose focal length is 0.0500 m.

(a) Find the image distance and determine whether the image is real or virtual. (b) Find the magnification and height of the image on the film.

Example 9 The Real Image Formed by a Camera Lens

A 1.70-m tall person is standing 2.50 m in front of a camera. The camera uses a converging lens whose focal length is 0.0500 m.

(a) Find the image distance and determine whether the image is real or virtual. (b) Find the magnification and height of the image on the film.

$$(a) \quad \frac{1}{d_i} = \frac{1}{f} - \frac{1}{d_o} = \frac{1}{0.0500 \text{ m}} - \frac{1}{2.50 \text{ m}} = 19.6 \text{ m}^{-1}$$

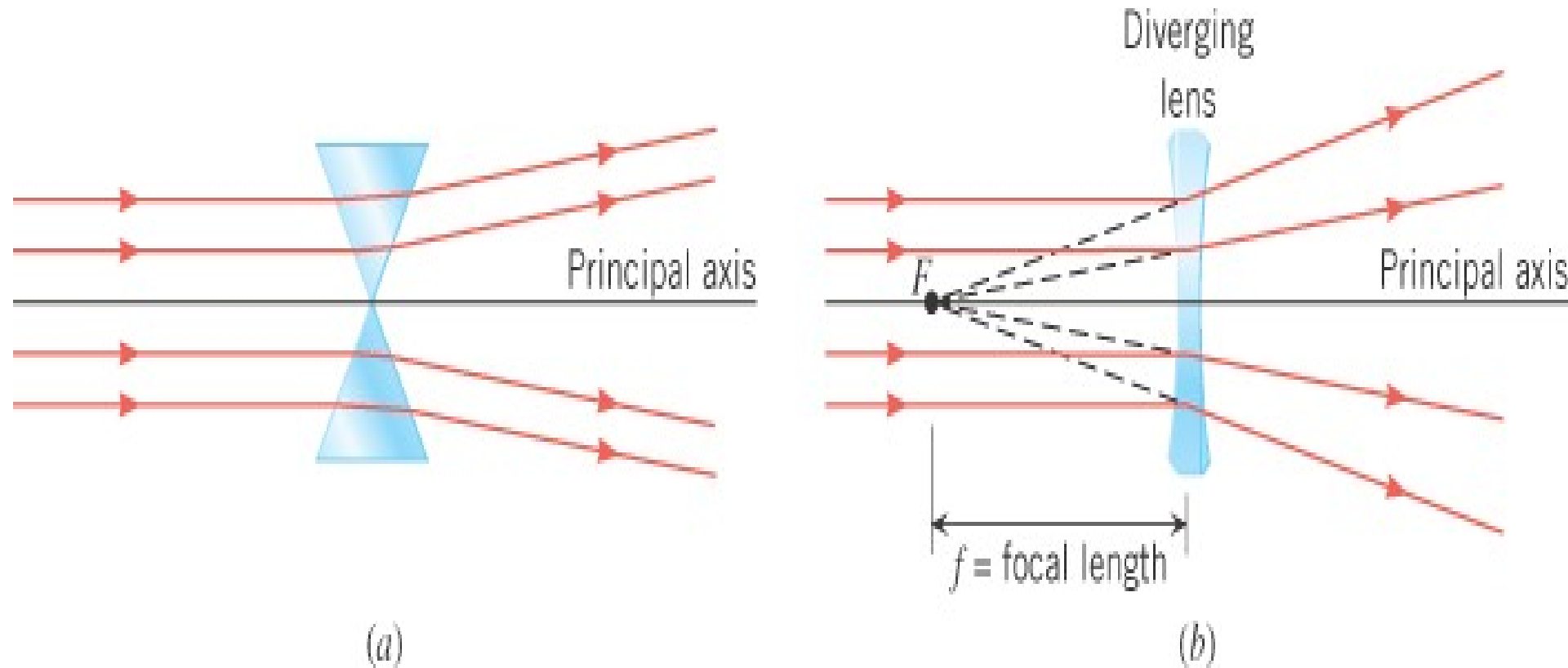
$$d_i = 0.0510 \text{ m} \quad \text{real image}$$

$$(b) \quad m = -\frac{d_i}{d_o} = -\frac{0.0510 \text{ m}}{2.50 \text{ m}} = -0.0204$$

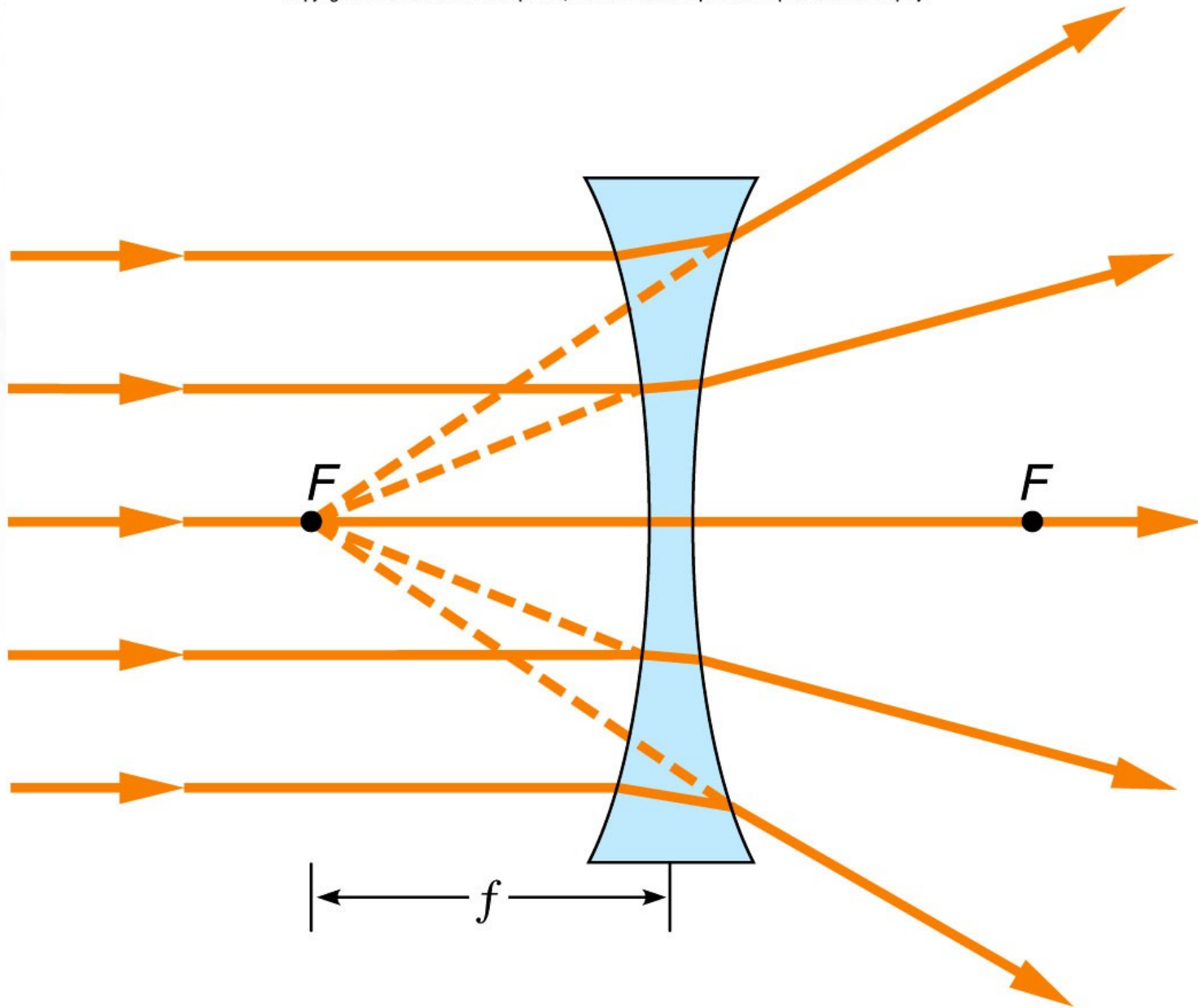
$$h_i = mh_o = (-0.0204)(2.50 \text{ m}) = -0.0347 \text{ m}$$

26.6 Lenses

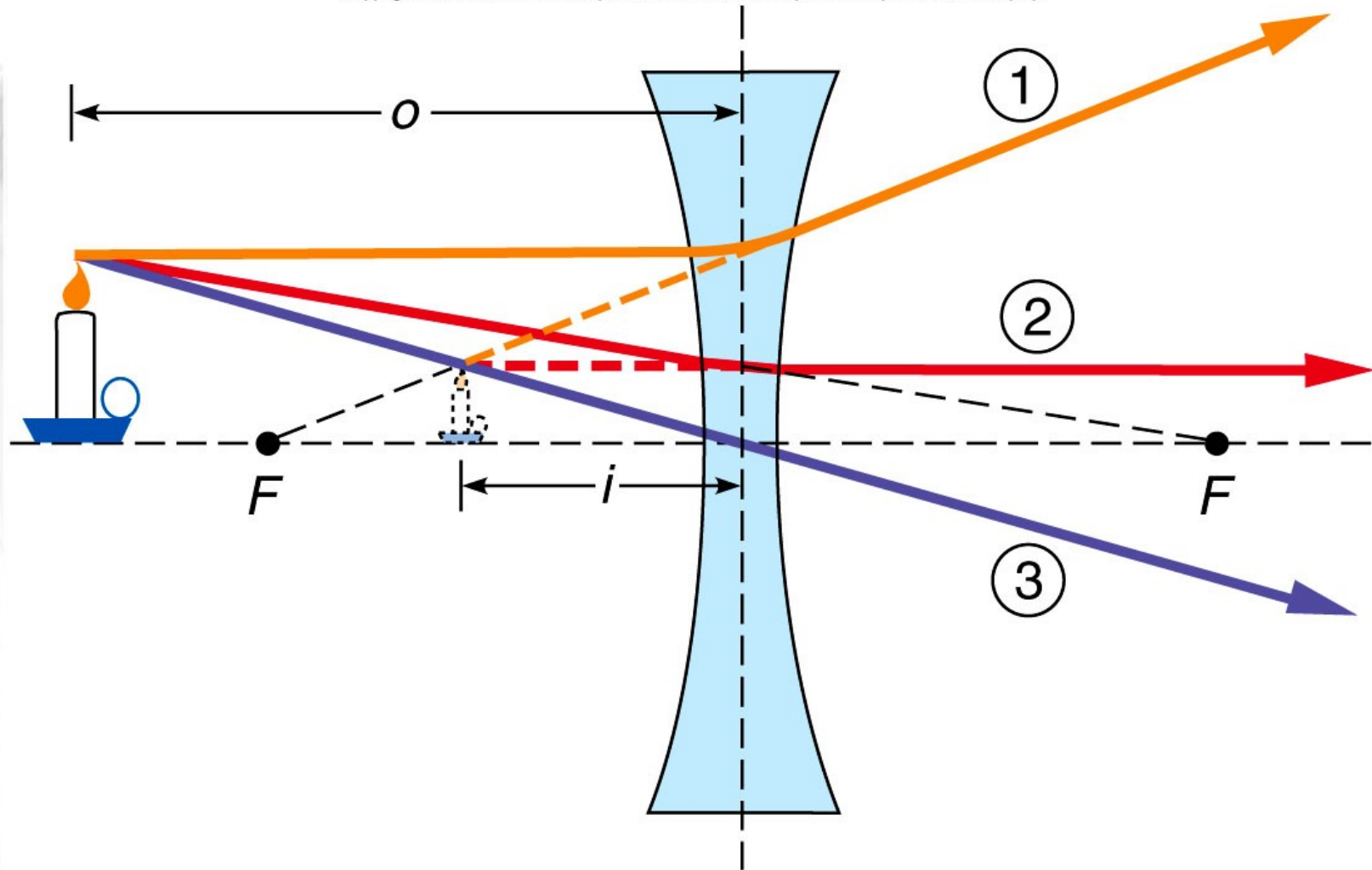
With a diverging (concave) lens, paraxial rays that are parallel to the principal axis appear to originate from the focal point.



Glasses or shortsighted = concave lenses.



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Summary of Sign Conventions for Lenses

f is + for a converging lens.

f is – for a diverging lens.

d_o is + if the object is to the left of the lens.

d_o is – if the object is to the right of the lens.

d_i is + for an image formed to the right of the lens (real image).

d_i is – for an image formed to the left of the lens (virtual image).

m is + for an upright image.

m is – for an inverted image.

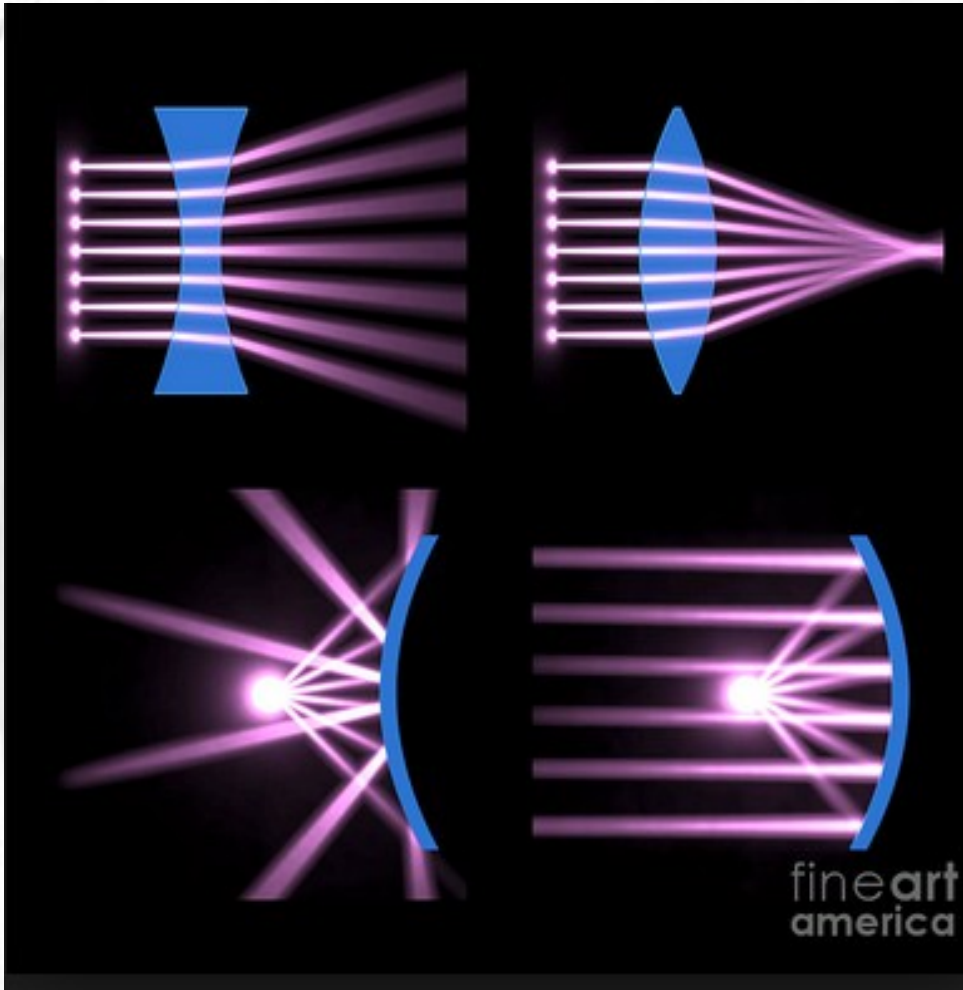
How far from the lens is the image formed of a lamp 35 centimeters away from a lens of focal length + 20 centimeters?

How far from the lens is the image formed by a + 20 centimeters lens if the object is 12 centimeters from it?

What focal length lens is needed to form a virtual image 12 centimeters from the lens when the object is 35 centimeters from the lens?

Find the size of the image of a book 18 centimeters high placed 30 centimeters from a diverging lens of focal length -20 centimeters.

https://www.youtube.com/watch?v=yMO7Lsjhmi4&list=PLX2gX-ftPVXWA5TjEhVQSQQzZ-5_5Nui8



Identify :

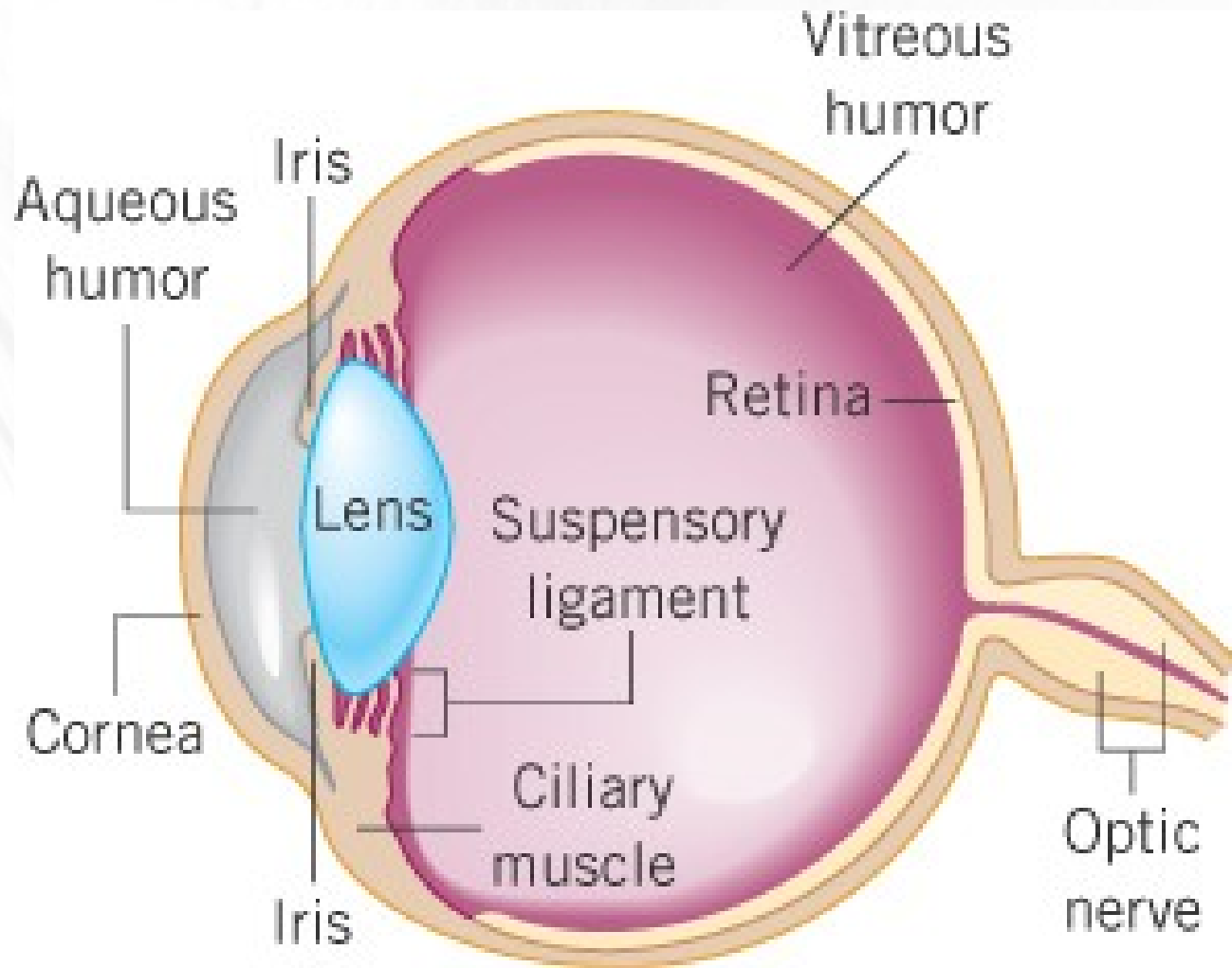
Convex lens, concave lens,
Convex mirror, concave mirror

Which ones make rays
Converge ?

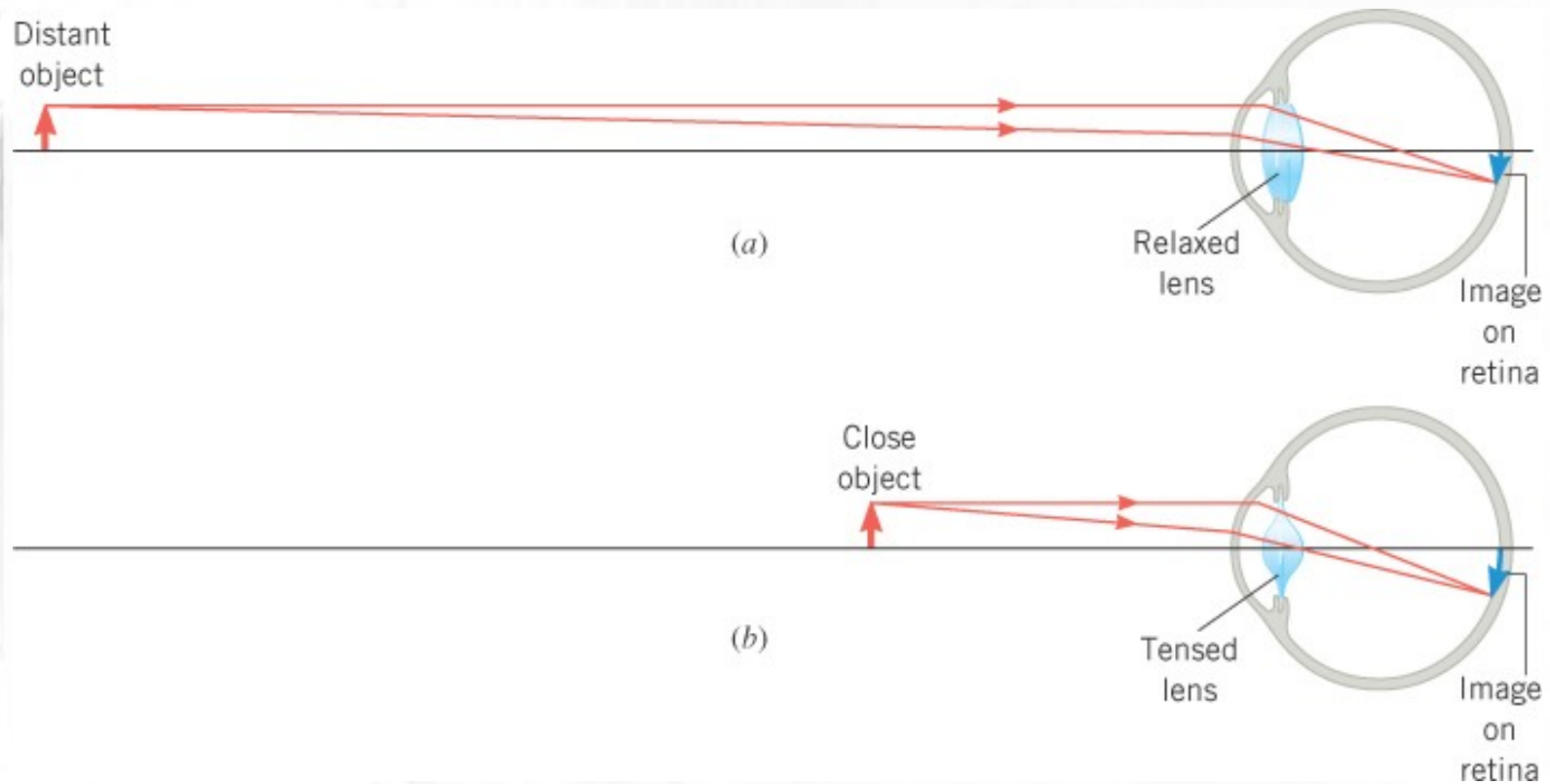
Diverge ?

Run simulation (including mirror)

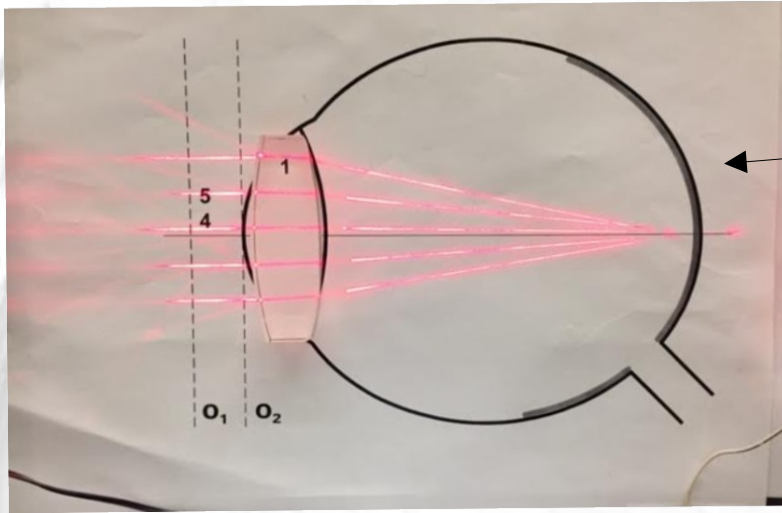
Show pdf telescopes



OPTICS

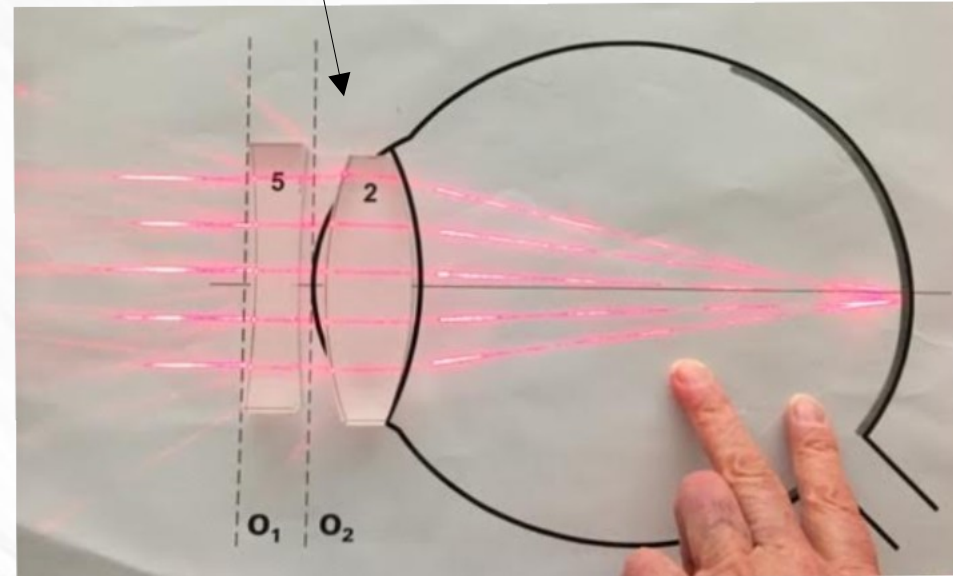
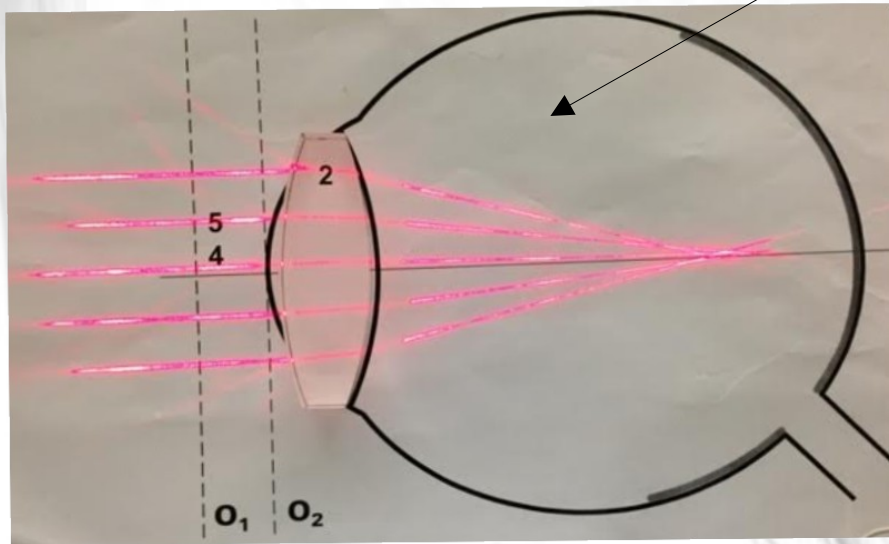


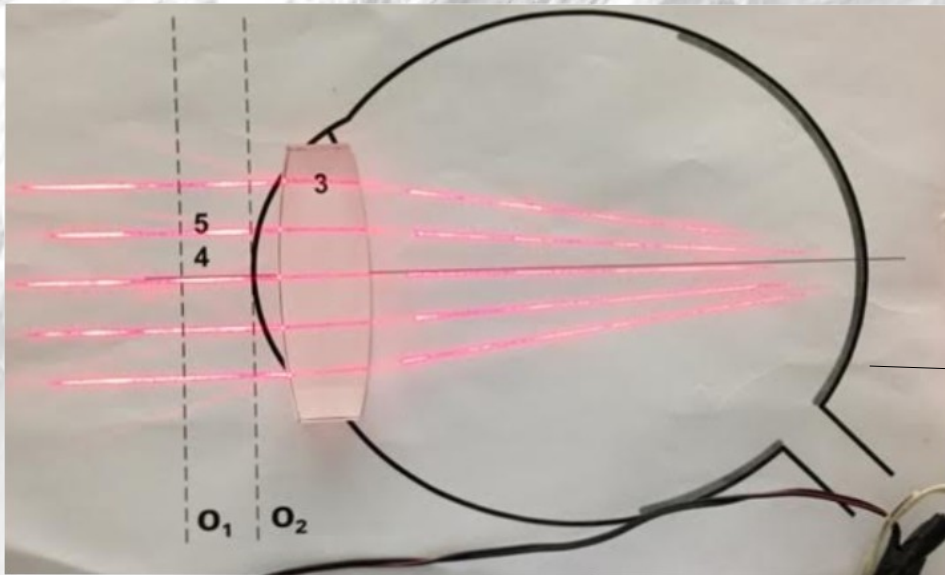
The lens only contributes about 20-25% of the refraction, but its function is important.



Normal eye

Eye too "flat" → near sightness →
Needs diverging lenses

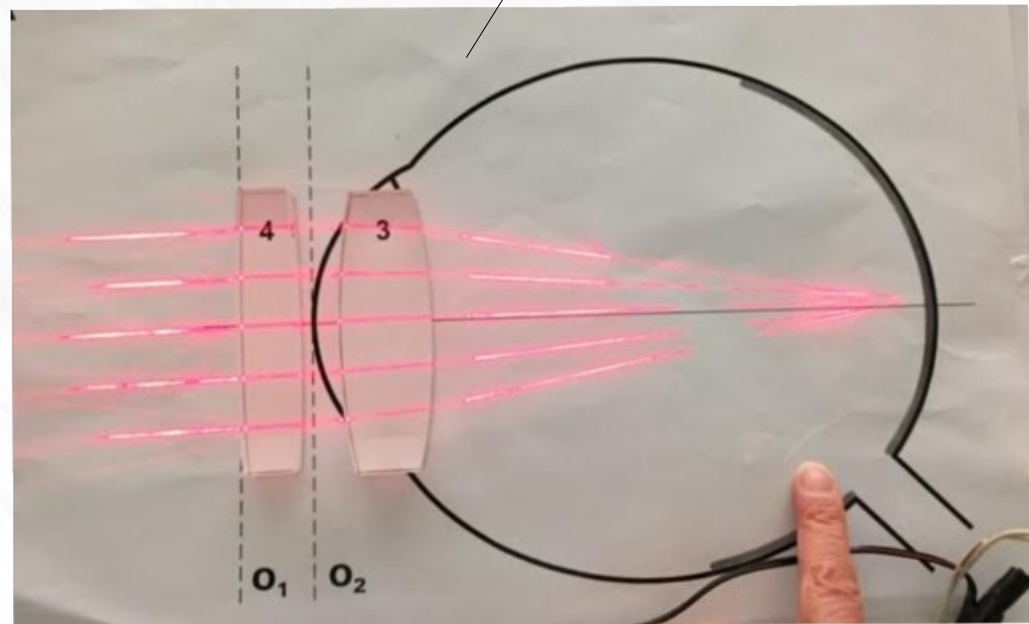




Eye too round or
Can't contract the "lens" anymore
(oldness ..)

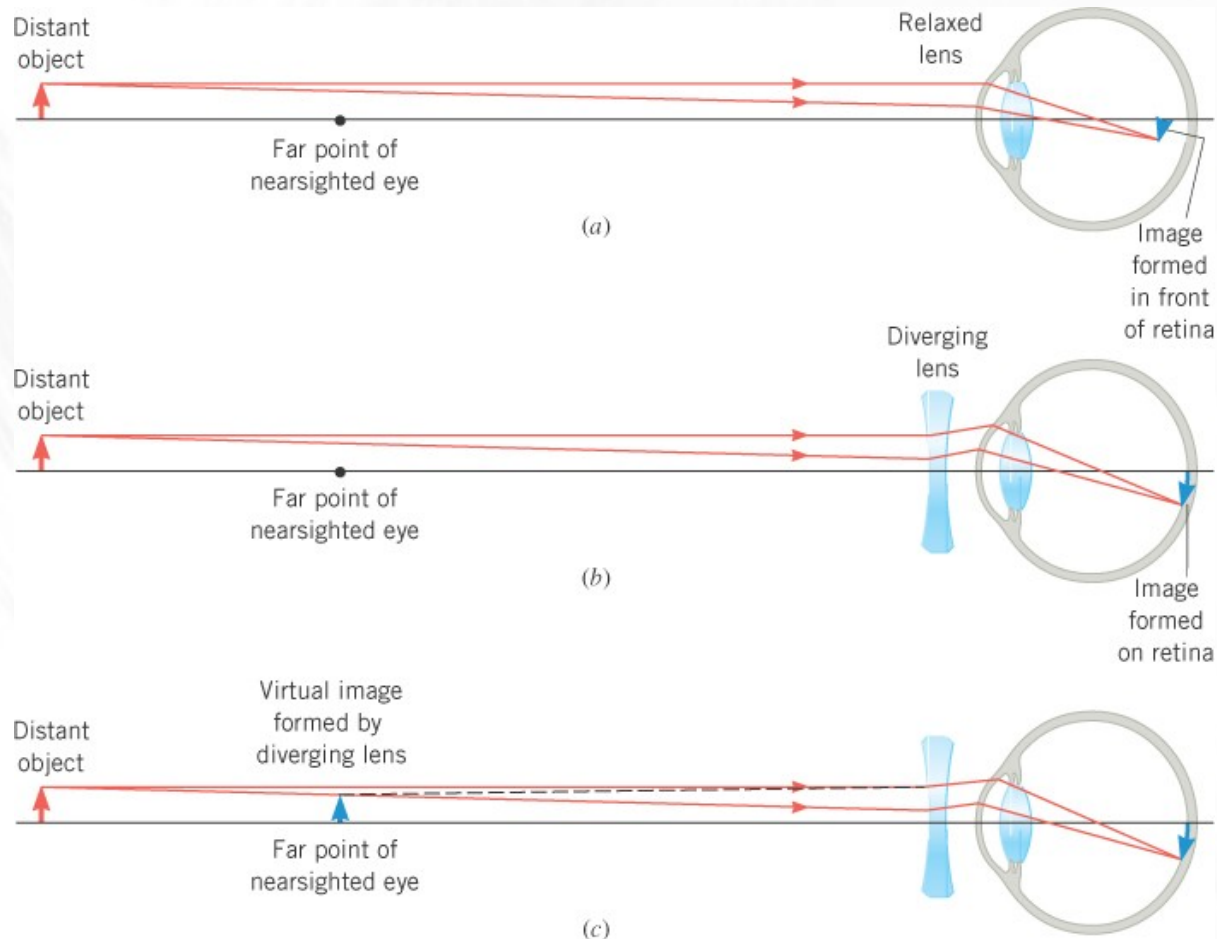
→ far sightness →

Needs converging lenses



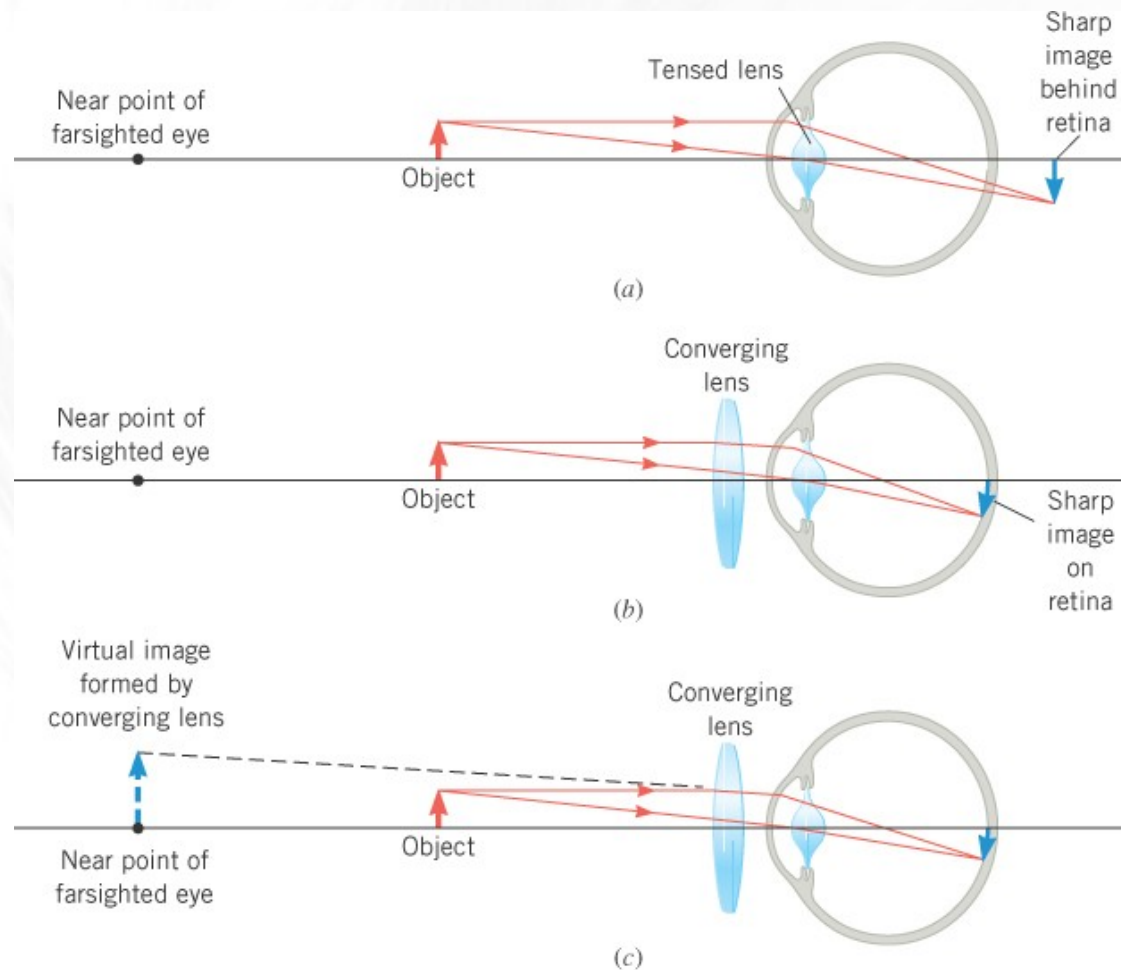
NEARSIGHTEDNESS

https://javalab.org/en/correction_of_near_sightedness_en/



The lens creates an image of the distance object at the far point of the nearsighted eye.

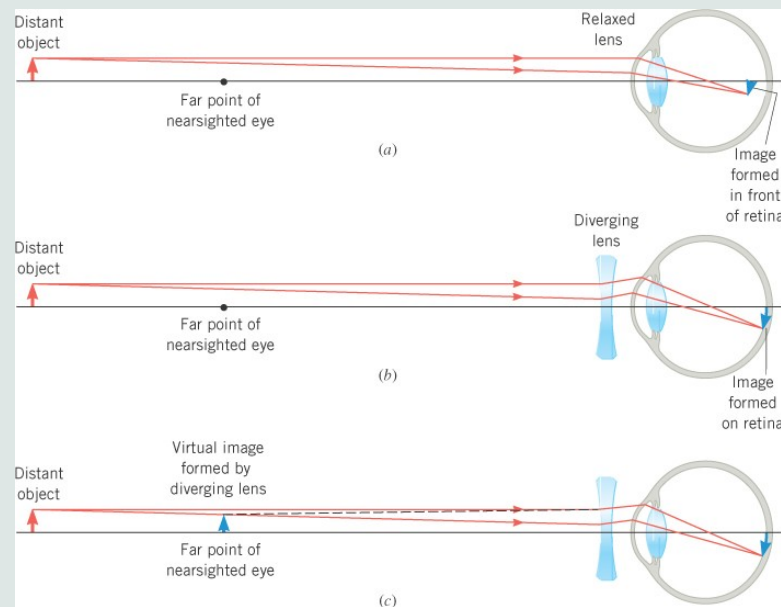
FARSIGHTEDNESS



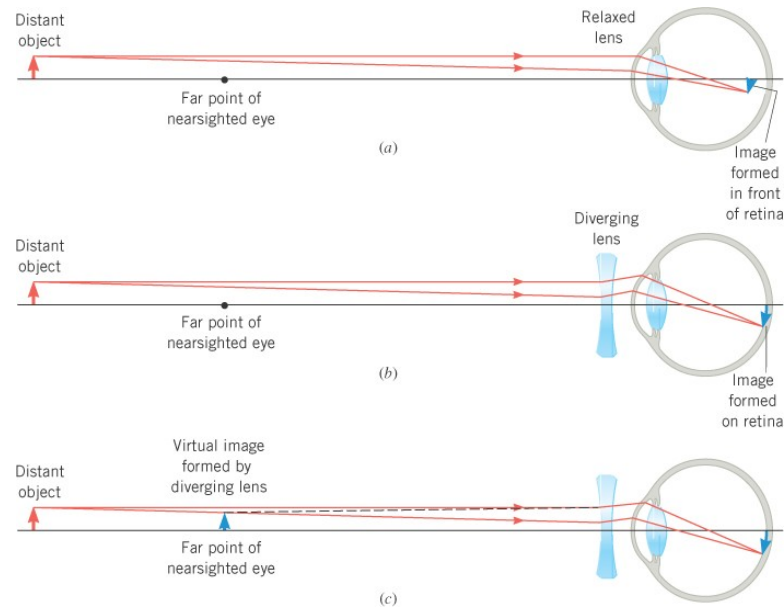
The lens creates an image of the close object at the near point of the farsighted eye.

Example 12 Eyeglasses for the Nearsighted Person

A nearsighted person has a far point located only 521 cm from the eye. Assuming that eyeglasses are to be worn 2 cm in front of the eye, find the focal length needed for the diverging lens of the glasses so the person can see distant objects.



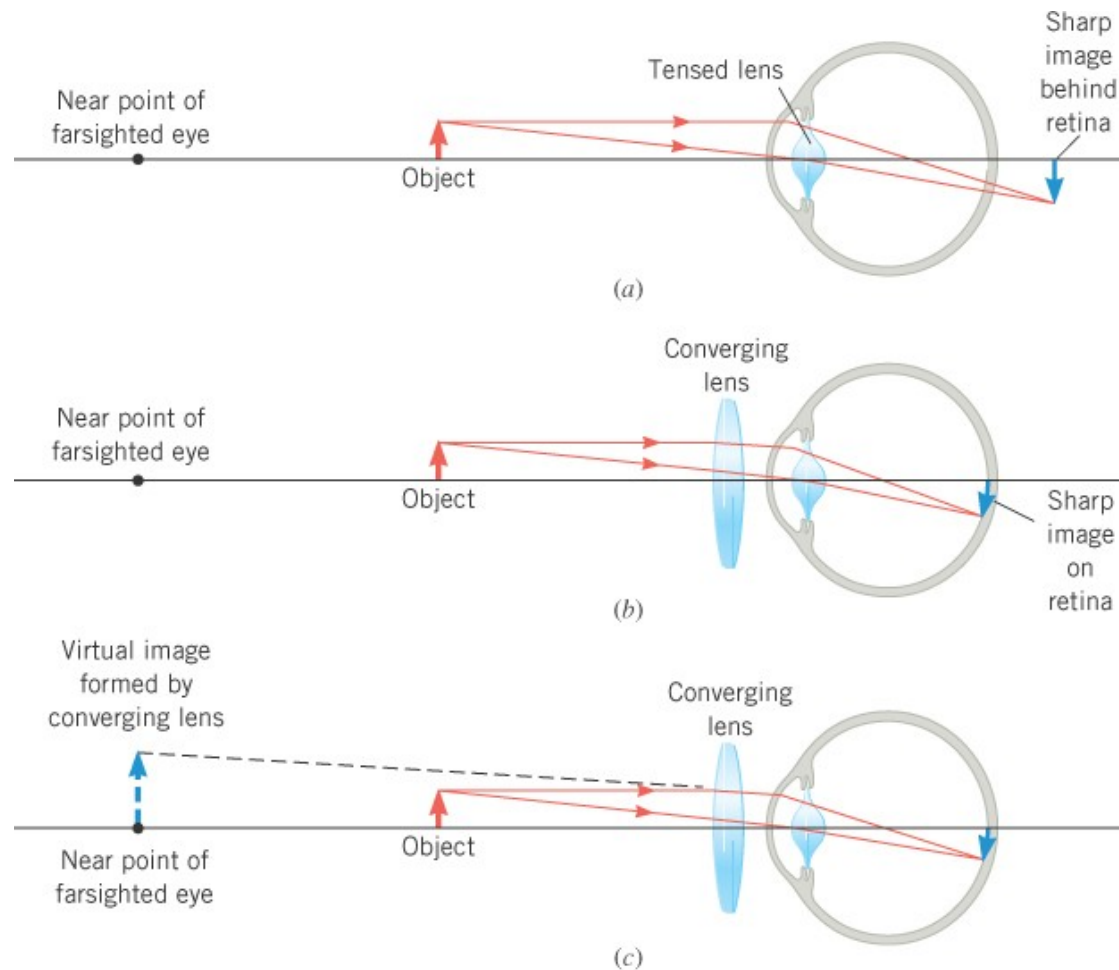
26.10 The Human Eye



$$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i} = \frac{1}{\infty} - \frac{1}{519 \text{ cm}}$$

$$f = -519 \text{ cm}$$

FARSIGHTEDNESS



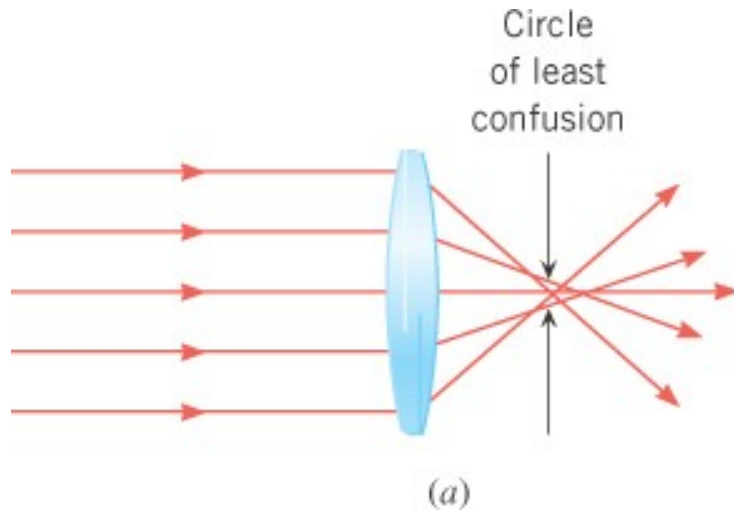
The lens creates an image of the close object at the near point of the farsighted eye.

THE REFRACTIVE POWER OF A LENS – THE DIOPTER

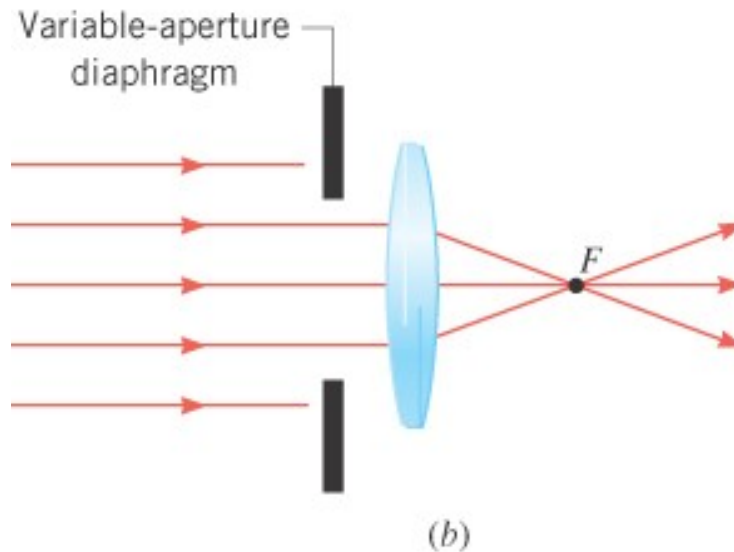
Optometrists who prescribe correctional lenses and the opticians who make the lenses do not specify the focal length. Instead they use the concept of ***refractive power***.

$$\text{Refractive power (in diopters)} = \frac{1}{f \text{ (in meters)}}$$

26.14 Lens Aberrations

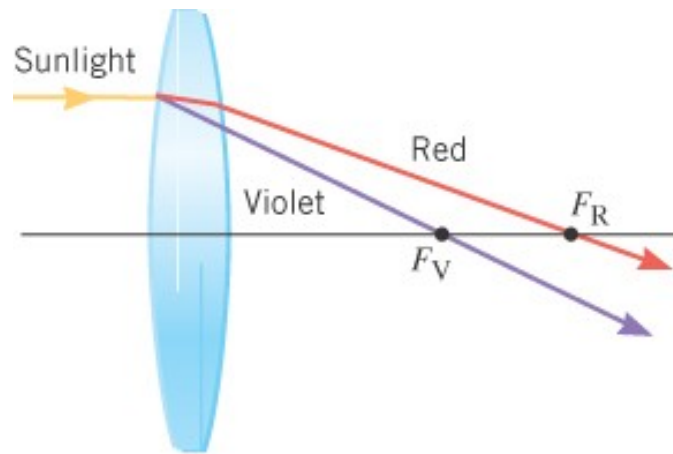


In a converging lens, spherical aberration prevents light rays parallel to the principal axis from converging at a single point.

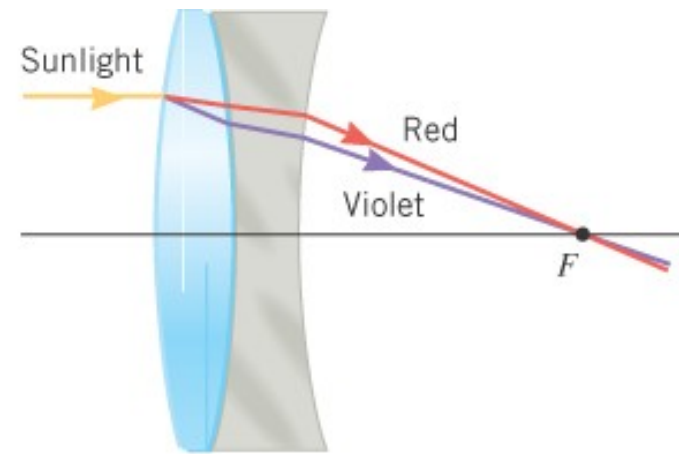


Spherical aberration can be reduced by using a variable-aperture diaphragm.

26.14 Lens Aberrations



(a)



(b)

Chromatic aberration arises when different colors are focused at different points along the principal axis.